

Heterogeneous Households and Aggregate Dynamics: Forecast-Invariance and Data Inversion¹

Online Appendix

Contents

A Proofs	3
A.1 Proof of Proposition 1	3
A.2 Proof of Proposition 2	4
A.3 Proof of Proposition 3	4
A.4 Proof of Corollary 2	5
A.5 Proof of Corollaries 3 and 4	6
A.6 Proof of Theorem 1	6
A.7 Proof of Proposition 4	6
A.8 Saving share under time-varying β	6
B Extended Theoretical Discussion	8
B.1 Transition matrix interpretation	8
B.2 Signal-to-noise ratios	8
B.3 Special cases: extended discussion	8
B.4 State-dependent inequality dynamics	9
C Asset Pricing Details	9
C.1 Static and dynamic decomposition	10
C.2 Wealth mobility	10
C.3 Risk premium sensitivity	11
D Epstein-Zin Extension	11
E Additional Quantitative Results	12
E.1 Implied TFP comparison	12
E.2 Structural interpretation of ZL	13
E.3 Aggregate dynamics	15
E.4 Counterfactual multi-variable panels	17

¹Minji Bang, University of Cambridge, Email: mb2503@cam.ac.uk; Hanbaek Lee, University of Cambridge, Email: hl610@cam.ac.uk

E.5	Transition dynamics	18
E.6	Distributional state dynamics	18
E.7	Consumption dynamics	19
E.8	Historical episode analysis	20
E.9	Cross-sectional distributional dynamics	20
E.10	Robustness	21
F	Micro inversion: data construction	22
F.1	Common conventions	23
F.2	PSID biennial saving-equation derivation	23
F.3	PSID variables and sample	24
F.4	SCF variables and sample	25
F.5	beta-clipping and quality flags (SCF only)	26
F.6	Output files and reproducibility	26
G	Theory: definitions, proofs, and implications	27
G.1	Existence and characterization of the MRCE	28
G.2	Forecast-invariant equilibrium	29
G.3	Closure of the moment recursion	29
G.4	Ergodicity	30
G.5	Special cases	30
G.6	Macroeconomic implications	31
G.7	Asset pricing implications	32
G.8	Extension: a household-level leisure weight θ_i	32
H	Supplementary empirical results	33
I	Additional results for Section 5	33
I.1	Sorting accelerator: decomposition, incidence, and cross-sectional dynamics	34
I.2	Distributional impact of capital-tax increases	39
I.3	Household parameters: quartile and age-conditional detail	39

A Proofs

A.1 Proof of Proposition 1

Proposition 1 (Analytical household policies and wealth dynamics).

The optimal consumption and labor supply policies are:

$$c(x; X) = \frac{1 - \beta}{1 + \theta} (1 + r(X)(1 - \tau)e)a + \left(\frac{1 - \beta}{1 + \theta} a\xi + \kappa \right) w(X)(1 - \tau)Z_L z \quad (1)$$

$$n(x; X) = \max \left\{ \kappa(1 + \theta) + \xi a - \frac{\theta c(x; X)}{w(X)(1 - \tau)Z_L z}, 0 \right\} \quad (2)$$

The household's chosen saving is $\hat{a}'(x; X) = \beta a ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi)$.

The actual law of motion for wealth, after the unanticipated government transfer, is:

$$a'(x; X) = \hat{a}'(x; X) + T(X) = \beta a ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi) + T(X). \quad (3)$$

Proof.

I assume the constraint $a' \geq 0$ does not bind and later verify that the solution satisfies this. The household is unaware of the transfer T and optimizes based solely on factor income. From the first-order conditions,

$$[c] : \frac{1}{c - \kappa w(X)(1 - \tau)Z_L z} = \lambda \quad (4)$$

$$[a'] : \beta \mathbb{E}[v_a(x'; X')] = \lambda \quad (5)$$

$$[n] : \frac{\theta}{\kappa + \xi a - n} = \lambda w(X)(1 - \tau)Z_L z. \quad (6)$$

From the envelope condition,

$$[a] : v_a(x; X) = \frac{\theta \xi}{\kappa + \xi a - n} + \lambda(1 + r(X)(1 - \tau)e). \quad (7)$$

From Equations (4) and (6),

$$\frac{\theta}{\kappa + \xi a - n} = \frac{w(X)(1 - \tau)Z_L z}{c - \kappa w(X)(1 - \tau)Z_L z}. \quad (8)$$

Rearranging,

$$n(x; X) = \kappa(1 + \theta) + \xi a - \theta \frac{c(x; X)}{w(X)(1 - \tau)Z_L z}. \quad (9)$$

The budget constraint can be written as:

$$(1 + \theta)(c - \kappa w(X)(1 - \tau)Z_L z) + a' = ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z \xi)a. \quad (10)$$

Conjecture $v(a, z, e; X) = A(z, e, X) \log a + B(z, e, X)$. Under log utility, the Euler equation combined with the conjectured form yields $a' = \beta a R_i$, where $R_i = (1 + \tilde{r}e + \tilde{w}Z_L z \xi)$ —the standard result of [Brock and Mirman \(1972\)](#). The government then adds the unanticipated transfer: $a'_{\text{actual}} = \beta a R_i + T$. Substituting back confirms $A(z, e, X) = (1 + \theta)/(1 - \beta)$, and B satisfies a well-defined recursion. The coefficient A is independent of beliefs about future aggregates, which is the basis for Theorem 1; see Appendix [A.8](#) for the general ζ recursion when β varies. Finally, the household's chosen saving $\hat{a}' = \beta a R_i$ is non-negative for every $a \geq 0$, since $\beta \in (0, 1)$ and $R_i \geq 0$ under the maintained assumption $1 + \tilde{r}e \geq 0$; so the borrowing constraint $a' \geq 0$ is slack in the optimization itself, independent of T . ■

A.2 Proof of Proposition 2

Proof. Write $s(a) = (\alpha a + T)/(\gamma a + K_d + T)$, where

$$\alpha \equiv \beta(1 + \tilde{r}e + \tilde{w}Z_L z \xi) - 1, \quad \gamma \equiv \frac{\tilde{w}Z_L z \xi(1 + \theta\beta)}{1 + \theta} - \frac{\theta(1 - \beta)(1 + \tilde{r}e)}{1 + \theta} + \tilde{r}e, \quad K_d \equiv \tilde{w}Z_L z \kappa \theta.$$

Property (ii): $s(0; X) = T/(K_d + T) > 0$ since $T > 0$ and $K_d > 0$. Property (i): $\lim_{a \rightarrow \infty} s(x; X) = \alpha/\gamma$, the limit of the fractional-linear function in a . Property (iii): direct differentiation gives

$$\frac{\partial s}{\partial a} = \frac{\alpha(K_d + T) - \gamma T}{(\gamma a + K_d + T)^2}, \quad \frac{\partial^2 s}{\partial a^2} = \frac{-2\gamma[\alpha(K_d + T) - \gamma T]}{(\gamma a + K_d + T)^3}.$$

The numerator $\Delta \equiv \alpha(K_d + T) - \gamma T$ is positive when two conditions hold: (a) $\alpha > 0$, which is the dynamic-efficiency condition $\beta R_i > 1$; and (b) K_d dominates the transfer correction $-(\gamma - \alpha)T$. Both conditions are satisfied throughout the parameter region examined in the quantitative section. Specifically, $\beta R_i > 1$ in every period of the historical inversion, since $\tilde{r} + \tilde{w}Z_L \xi > (1 - \beta)/\beta \approx 0.04$. The transfer is small relative to subsistence: $T \approx 0.001K \ll K_d$ at the calibrated η . Hence $\Delta > 0$ and $\gamma > 0$, giving $\partial s/\partial a > 0$ and $\partial^2 s/\partial a^2 < 0$. ■

A.3 Proof of Proposition 3

Proof. The saving policy $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L \xi z) + T$ is linear in a with coefficients that depend on (z, e) and current prices. Aggregating across the distribution:

For aggregate capital: $K' = \int a'(x; X) d\Phi = \beta[(1+\tilde{r}+\tilde{w}Z_L\xi)K + \tilde{r}\text{cov}(e, a) + \tilde{w}Z_L\xi\text{cov}(z, a)] + T$, using $\mathbb{E}[e] = \mathbb{E}[z] = 1$.

For part (i) (contemporaneous sufficiency): Proposition 1 shows that individual policies $c(x; X)$, $a'(x; X)$, $n(x; X)$ are linear in a with coefficients depending on (z, e) and prices (r, w) . The labor supply equation depends on Φ only through $(K, \text{cov}(z, a), \text{cov}(e, a))$, so prices are functions of $\mathcal{M}$ alone. Hence all equilibrium allocations are determined by $\mathcal{M}$ and the aggregate state.

For part (iii) (factor prices and aggregate labor): direct aggregation of the individual policies using $\mathbb{E}[e] = \mathbb{E}[z] = 1$ yields the stated expressions.

Unique existence of aggregate labor. The aggregate labor equation (20) is implicit in L because the after-tax wage $\tilde{w} = (1 - \tau_L)(1 - \alpha)A(K_{eff}/L)^\alpha$ depends on L . Define $g(L) \equiv L - \text{RHS}(L)$ where RHS is the right-hand side of (20) evaluated at prices consistent with L . The first term $(\kappa + Z_L\xi \cdot f(\beta, \theta) \cdot (K + \text{cov}(z, a)))$ is independent of L . The second term, $-\theta(1-\beta)/((1+\theta)\tilde{w}Z_L) \cdot W$ where $W = K(1+\tilde{r}) + \tilde{r}\text{cov}(e, a)$, depends on L through $\tilde{w} \propto L^{-\alpha}$ and $\tilde{r} \propto L^{1-\alpha}$. Since $\alpha \in (0, 1)$, $\tilde{w}$ is strictly decreasing in L while $W/\tilde{w} \propto L^\alpha \cdot L^{1-\alpha} = L$. Hence the negative term scales as $-c \cdot L$ for some $c > 0$, making $g'(L) = 1 + c > 0$ strictly positive. Continuity and strict monotonicity of g guarantee a unique zero. ■

Remark on distributional dynamics. For $\text{cov}(z', a')$: expanding $\mathbb{E}[z'a'] - \mathbb{E}[z']\mathbb{E}[a']$ using $z' = (1 - \rho_z) + \rho_z z + \varepsilon'_z$ and $a' = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L\xi z) + T$ yields terms involving $\mathbb{E}[\hat{z}^2 a]$ and $\mathbb{E}[\hat{z}\hat{e}a]$ (where $\hat{z} = z - 1$, $\hat{e} = e - 1$). These three-way cross-moments are not, in general, functions of $\mathcal{M}$ alone. Under joint normality of $(\hat{z}, \hat{e}, a)$, the Isserlis (Wick) theorem gives $\mathbb{E}[\hat{z}^2 a] = \sigma_z^2 K$ and $\mathbb{E}[\hat{z}\hat{e}a] = 0$ (using independence of z and e), and the covariance dynamics close into a 3×3 linear recursion. However, the Kesten-process wealth dynamics generate Pareto tails that violate normality. Therefore, tracking the full distribution via agent-based simulation is necessary for the forward evolution of the covariances, while the K' equation and all contemporaneous allocations remain exact. ■

A.4 Proof of Corollary 2

Proof. The individual law of motion $a' = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L\xi z) + T$ is a Kesten process with multiplicative coefficient $R_i = \beta((1 + \tilde{r}e) + \tilde{w}Z_L\xi z)$ and additive intercept $T > 0$. By Kesten (1973), when $\mathbb{E}[\log R_i] < 0$ and $T > 0$, the process admits a unique stationary distribution with Pareto tails: $\Pr(a > x) \sim x^{-\zeta}$ for large x , where ζ solves $\mathbb{E}[R_i^\zeta] = 1$. The additive component $T > 0$ is essential: without it, the purely multiplicative system does not admit a non-degenerate stationary solution. ■

A.5 Proof of Corollaries 3 and 4

Proof of Corollary 3. Set $e = 1$ throughout. Then $\text{cov}(e, a) = 0$ for all t , and the K' equation reduces to $K' = \beta[(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi \text{cov}(z, a)] + T$, with $\mathcal{M} = [K, \text{cov}(z, a)]$ sufficient for all contemporaneous allocations. ■

Proof of Corollary 4. Set $e = 1$ and $z = 1$. Then both covariances vanish identically, and the system reduces to a single equation $K' = \beta(1 + \tilde{r} + \tilde{w}Z_L\xi)K + T$. The aggregate consumption function follows from $C = Y + (1 - \delta)K - K'$. ■

A.6 Proof of Theorem 1

Proof. The saving policy $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L\xi z) + T$ depends only on current after-tax prices $(\tilde{r}, \tilde{w})$, current Z_L , the current transfer T , and the household's current state (a, z, e) . None of these objects involves expectations of future aggregate states. The consumption policy $c(x; X)$ and labor supply $n(x; X)$ inherit this property. Since all individual policies are independent of beliefs about future aggregates, the aggregate law of motion—including K' from the K' equation in the main text (eq. (17)) and the distributional evolution of Φ —is also independent of beliefs: it depends only on current prices (which are functions of the current state $\mathcal{M}$) and the current transfer.

The mechanism is the certainty-equivalence property of log preferences: income and substitution effects of future return changes exactly cancel, so saving depends only on current resources and prices. This aggregates because linear policies preserve the moment structure. ■

A.7 Proof of Proposition 4

Proof. The FRCE requires that household policies be optimal given *arbitrary* beliefs about aggregate dynamics. Theorem 1 establishes that the saving policy is forecast-invariant. Since the aggregate law of motion depends only on the collection of saving policies (through the moment aggregation), and since these policies are identical regardless of beliefs, the FRCE law of motion coincides with the MRCE law of motion. Uniqueness of the MRCE then implies uniqueness of the FRCE, and the allocations are identical. ■

A.8 Saving share under time-varying β

Derivation. Since β is an element of the aggregate state X and may vary over time, the Bellman equation for the multiplicative component of wealth (ignoring the additive transfer

T) is:

$$v_t(a, z, e; X) = \max_{a'} (1 + \theta) \log \tilde{c} + \beta_t \mathbb{E}_t [v_{t+1}(a', z', e'; X')], \quad (11)$$

where $\tilde{c} \equiv c - \kappa w(1 - \tau)Z_L z$ is consumption net of the subsistence floor. Conjecture $v_t(a, z, e; X) = \zeta_t \log a + B_t(z, e, X)$, where ζ_t may depend on aggregate conditions but not on individual wealth.

From the envelope condition, $v_{a,t} = \zeta_t/a$. Combined with the first-order conditions (as in Appendix A.1), the envelope condition yields $\zeta_t/a = \lambda_t \Pi_t$, where $\Pi_t = (1 + \tilde{r}_t e) + \tilde{w}_t Z_L \xi z$ is the gross multiplicative return on wealth. Since $\lambda_t = 1/\tilde{c}_t$, this gives $\tilde{c}_t = a \Pi_t / \zeta_t$, and the budget constraint implies:

$$a' = \Pi_t a \left(1 - \frac{1 + \theta}{\zeta_t}\right). \quad (12)$$

The first-order condition for a' requires $\beta_t \mathbb{E}_t [\zeta_{t+1}/a'] = \zeta_t/(a \Pi_t)$. Substituting the expression for a' and simplifying:

$$\zeta_t = (1 + \theta) + \beta_t \mathbb{E}_t [\zeta_{t+1}]. \quad (13)$$

The saving share out of the multiplicative component is therefore:

$$s_t = 1 - \frac{1 + \theta}{\zeta_t} = \frac{\beta_t \mathbb{E}_t [\zeta_{t+1}]}{\zeta_t}. \quad (14)$$

Constant β . When β is constant, $\zeta = (1 + \theta)/(1 - \beta)$ and $s = \beta$, recovering the steady-state result.

Forecast invariance. Under the FRCE, households treat the current β as permanent. The recursion (13) then gives $\zeta = (1 + \theta)/(1 - \beta)$, so $s = \beta$. This is a static function of the current discount factor—no expectations are required. The saving policy is $a'(x; X) = \beta a \Pi + T$, and the aggregate capital law of motion $K' = \beta[(1 + \tilde{r} + \tilde{w} Z_L \xi)K + \tilde{r} \text{cov}(e, a) + \tilde{w} Z_L \xi \text{cov}(z, a)] + T$ inherits this forecast invariance.

Time-varying τ_t . The tax rate enters only through after-tax prices in the income base Π_t , and through the transfer T_t (which is an MIT shock). Since τ_t does not appear in the ζ recursion (13), time-varying flat tax rates preserve the analytical structure without any additional state variables.

Separate capital and labor tax rates. The same logic extends to separate time-varying tax rates $\tau_{K,t}$ and $\tau_{L,t}$. After-tax factor prices become $\tilde{r}_t = r_t(1 - \tau_{K,t})$ and $\tilde{w}_t = w_t(1 - \tau_{L,t})$, and the transfer is $T_t = \eta[\tau_{K,t}(r_t + \delta)K_t + \tau_{L,t} w_t L_t]$. Because both rates

enter multiplicatively on factor incomes, the household's next-period wealth $a' = s_t \cdot a\Pi_t + T_t$ remains *linear* in individual wealth a —the key property that preserves the analytical structure. In particular, the saving share $s_t = \beta_t$ is still independent of $(\tau_{K,t}, \tau_{L,t})$: tax rates shift *what* is saved (through Π_t and T_t) but not the *share* of wealth saved. Progressive taxes or tax brackets, by contrast, would make the effective marginal rate depend on household-level income $a\Pi_t$, breaking linearity in a and destroying the MRCE's closed-form aggregation.

■

B Extended Theoretical Discussion

B.1 Transition matrix interpretation

The MRCE state vector has three components: aggregate capital K (the mean of the wealth distribution), $\text{cov}(z, a)$ (the alignment between labor productivity and wealth), and $\text{cov}(e, a)$ (the alignment between return heterogeneity and wealth). The first is the single state that [Krusell and Smith \(1998\)](#) found nearly sufficient numerically.

The aggregate capital law of motion $K' = \beta[(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{r}\text{cov}(e, a) + \tilde{w}Z_L\xi\text{cov}(z, a)] + T$ shows how the two covariances interact with capital. The $\tilde{r}\text{cov}(e, a)$ term captures return-wealth sorting: high-return wealthy households save more than average returns and average wealth would predict. The $\tilde{w}Z_L\xi\text{cov}(z, a)$ term captures productivity-wealth sorting through wealth-leisure complementarity. The two sorting channels operate independently (because z and e are independent), connected only through their common dependence on K .

B.2 Signal-to-noise ratios

The signal-to-noise ratios $\phi_z \equiv \sigma_z^2/(\sigma_z^2 + \sigma_{\varepsilon_z}^2)$ and $\phi_e \equiv \sigma_e^2/(\sigma_e^2 + \sigma_{\varepsilon_e}^2)$ control the rate at which idiosyncratic shocks reshuffle households across the distribution. High ϕ (close to one) means persistent shocks and slow reshuffling: sorting between productivity and wealth strengthens over time. Low ϕ (close to zero) means rapid reshuffling: the covariance between productivity and wealth dissipates quickly. Economies with persistent idiosyncratic shocks exhibit slow-moving distributional states that interact strongly with aggregate fluctuations; economies with transitory shocks have fast-adjusting distributions that play a smaller aggregate role.

B.3 Special cases: extended discussion

Approximate aggregation When there is no return heterogeneity ($e = 1$), the system reduces to two states: K and $\text{cov}(z, a)$. This is the analytical counterpart of the Krusell–

Smith economy. [Krusell and Smith \(1998\)](#) found numerically that the mean of the wealth distribution is “nearly sufficient” to forecast prices. The present result shows that the mean is *exactly* sufficient if and only if $\text{cov}(z, a) = 0$. In practice, this covariance is small relative to K in many calibrations, explaining why approximate aggregation works well numerically. But the error grows with the persistence and dispersion of idiosyncratic productivity—precisely the conditions under which the Krusell–Smith approximation breaks down.

Representative-agent nesting When all heterogeneity is eliminated, both covariances vanish and the model reduces to a stochastic growth model with log preferences and endogenous labor. The aggregate consumption function $C = \frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \kappa\tilde{w}Z_L$ has a permanent-income term proportional to wealth and a subsistence term proportional to the wage. The heterogeneous-agent model adds distributional corrections that, while small in normal times, become significant during episodes of high inequality.

B.4 State-dependent inequality dynamics

The K' equation makes inequality feedback precise. Higher capital raises returns, which increases $\tilde{r} \text{cov}(e, a)$ (the return-sorting contribution to aggregate saving), which raises future capital further. Similarly, higher capital raises wages, which increases $\tilde{w}Z_L\xi \text{cov}(z, a)$ (the productivity-sorting contribution). This self-reinforcing loop generates endogenous amplification: a positive shock raises capital, which strengthens sorting, which raises future capital.

This “wealth spiral” is consistent with the secular rise in top wealth shares documented by [Piketty \(2014\)](#) and [Saez and Zucman \(2016\)](#). The rate of concentration is governed by the persistence of the distributional dynamics, measured by the dominant eigenvalue in the agent-based simulation. The steady-state level is pinned down by the balance between multiplicative compounding (R_i) and the additive transfer T . Two forces limit concentration. Consumption: $\beta < 1$ ensures that a fraction $1 - \beta$ of wealth is consumed each period. The transfer T provides an additive floor that benefits low-wealth households disproportionately.

C Asset Pricing Details

The model delivers a novel asset pricing result: individual return heterogeneity, traditionally viewed as diversifiable, carries a positive risk premium because it has aggregate consequences ([Constantinides and Duffie, 1996](#); [Storesletten et al., 2004](#)).

In the standard complete-markets framework, idiosyncratic risk washes out by the law

of large numbers. Here it does not, because return heterogeneity creates systematic wealth sorting: households with persistently high e accumulate more wealth, work less through the wealth effect, and thereby depress aggregate labor supply and returns for *everyone*. Because e affects aggregate outcomes through $\text{cov}(e, a)$, it is not fully diversifiable.

C.1 Static and dynamic decomposition

The asset pricing implications decompose into a static and a dynamic component.

Static effect Holding the aggregate state fixed, higher $\text{cov}(e, a)$ reduces the current return:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \text{cov}(e, a)} < 0 \quad (15)$$

Higher sorting strengthens the wealth effect on labor supply, reducing employment and the marginal product of capital.

Dynamic effect One period ahead, the effect is reinforced:

$$\frac{\partial r(\mathcal{M}', A')}{\partial \text{cov}(e, a)} = \frac{\partial r'}{\partial K'} \frac{\partial K'}{\partial \text{cov}(e, a)} + \frac{\partial r'}{\partial \text{cov}(e', a')} \frac{\partial \text{cov}(e', a')}{\partial \text{cov}(e, a)} < 0 \quad (16)$$

Both terms are negative: higher sorting today raises capital accumulation (depressing future returns through diminishing marginal products) and propagates into tomorrow's sorting (continuing to depress returns through labor supply).

C.2 Wealth mobility

The model provides an analytical half-life of wealth:

$$T(\mathcal{M}, A) = \frac{\log(2)}{\log(\beta(1 + r(\mathcal{M}, A)(1 - \tau)))} \quad (17)$$

Faster mean reversion of $\text{cov}(e, a)$ implies quicker distributional adjustment, but slow rank mobility means individual wealth positions change gradually. The combination generates persistent aggregate effects of idiosyncratic return shocks. These results suggest that models with heterogeneous returns (Fagereng et al., 2020; Bach et al., 2020) have implications for asset prices and aggregate dynamics, not just the wealth distribution.

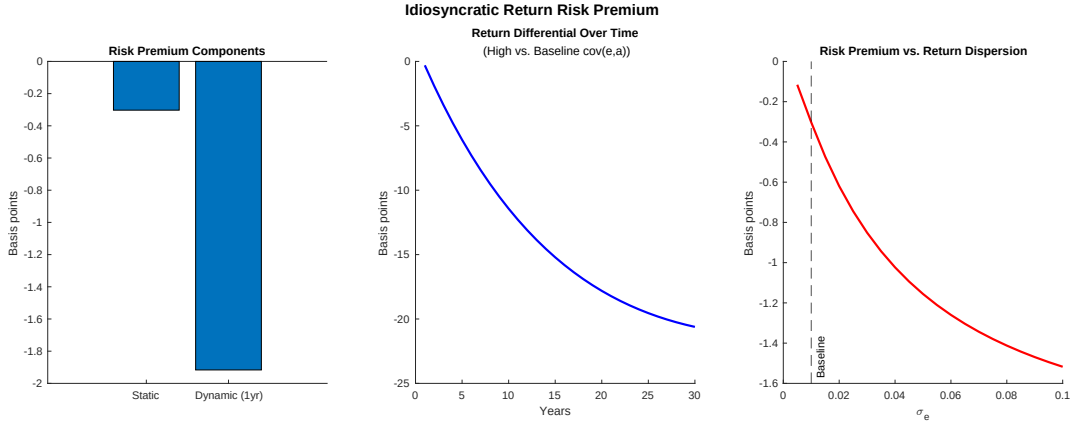
C.3 Risk premium sensitivity

To quantify the pricing channel, we compare steady-state economies at different σ_e . At baseline ($\sigma_e = 0.01$), the interest rate effect is -0.001 basis points—negligible. The premium grows quadratically:

$$\underbrace{-0.03 \text{ bp}}_{\sigma_e=0.05} \quad \underbrace{-0.10 \text{ bp}}_{\sigma_e=0.10} \quad \underbrace{-0.70 \text{ bp}}_{\sigma_e=0.50} \quad \underbrace{-1.01 \text{ bp}}_{\sigma_e=0.83}$$

The threshold for a 1 basis point premium requires $\sigma_e \approx 0.83$. At empirically realistic σ_e , the main consequence of return heterogeneity operates through the *amplification channel* (state-dependent impulse responses), not the *pricing channel*. Figure C.1 visualizes these results.

Figure C.1: Return risk premium sensitivity



Notes: Left: static vs. dynamic premium components at steady state. Center: cumulative return differential over 30 years. Right: premium as a function of return dispersion σ_e .

D Epstein-Zin Extension

Under Epstein-Zin preferences with unit elasticity of intertemporal substitution ($EIS = 1$), the flow composite is:

$$\tilde{c} \equiv c - \kappa Z_L z w(X)(1 - \tau), \quad m \equiv (\kappa + \xi a) - n, \quad X^{\text{comp}} \equiv \tilde{c} m^\theta.$$

The intratemporal optimality condition implies $m = (\theta/(w(X)(1 - \tau)Z_L z))\tilde{c}$, so:

$$X^{\text{comp}} = \tilde{c}^{1+\theta} \left(\frac{\theta}{w(X)(1 - \tau)Z_L z} \right)^\theta.$$

The recursive value function satisfies:

$$\log V_t = (1 - \beta) \log X_t + \beta \log (\mathbb{E}_t [V_{t+1}^{1-\gamma}])^{\frac{1}{1-\gamma}}, \quad \beta \in (0, 1), \quad \gamma > 0.$$

The critical result is that, with EIS = 1, the optimal saving share $s_a(s)$ satisfies the same type of fixed-point condition as in the log case:

$$s_a(s) = \left[1 + \left(\frac{\beta(1 + \theta)^p \bar{B}(s)}{C(s)} \right)^{\frac{1}{p-1}} \right]^{-1},$$

where $p \equiv (1 + \theta)(1 - \sigma)$, $C(s) \equiv (\theta/(w(X)(1 - \tau)Z_L z))^{\theta(1-\sigma)}$, and $\bar{B}(s) \equiv \mathbb{E}[B(s') \mid s]$ is a conditional expectation computed from the fixed-point system:

$$B(s) = \Pi(s)^p \left[C(s) \left(\frac{1 - s_a(s)}{1 + \theta} \right)^p + \beta s_a(s)^p \bar{B}(s) \right],$$

with $\Pi(s) \equiv (1 + r(X)(1 - \tau))e + w(X)(1 - \tau)Z_L z\xi$.

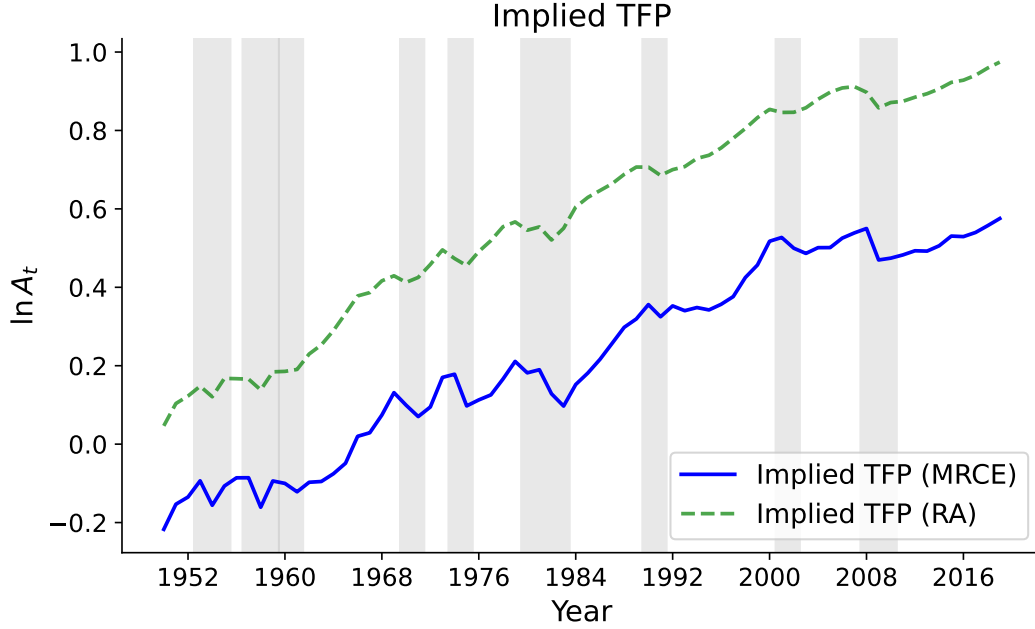
The MRCE and FRCE continue to hold under Epstein-Zin preferences with unit EIS. The saving share remains independent of wealth, and the law of motion retains its closed-form matrix structure. The parameter γ controls risk aversion over consumption *lotteries* (cross-sectional), while β and θ control intertemporal and intratemporal tradeoffs (time-series). This separation allows the model to match both the equity premium (through γ) and the saving rate (through β). With $\gamma \gg 1$, the wealth-sorting mechanism generates a substantially larger risk premium than under log utility.

E Additional Quantitative Results

E.1 Implied TFP comparison

Figure E.1 compares implied TFP from the MRCE (accounting for distributional covariances) with TFP from a representative-agent version (all covariances set to zero). The two series track closely at business cycle frequencies, but the HA model implies a slightly different level, reflecting distributional sorting's contribution to labor supply. The Fernald utilization-adjusted TFP series serves as an external benchmark.

Figure E.1: Implied TFP comparison



Notes: Implied TFP from the MRCE compared to the representative-agent analogue and the Fernald utilization-adjusted TFP series. Shaded areas indicate NBER recession dates.

E.2 Structural interpretation of $Z_{L,t}$

The aggregate labor-efficiency index $Z_{L,t}$, introduced in Section 2, is recovered from the labor-supply equation as a wedge, but it has a structural place in the model rather than being a free residual appended to it. Three properties distinguish it from a reduced-form labor wedge.

Model primitive with distributional consequences $Z_{L,t}$ scales household effective earnings $\tilde{w}_t Z_{L,t} z$ without entering the production function as a separate factor (it enters only through the efficiency-weighted aggregate $L_t \equiv Z_{L,t} \int z n d\Phi$, as discussed below). It appears in three places in the equilibrium:

1. The effective wage in household labor supply: $\tilde{w}_t Z_{L,t}$,
2. The wealth-leisure complementarity: $\tilde{w}_t Z_{L,t} \tilde{\xi}_t$, where $\tilde{\xi}_t \equiv \xi / A_t^{1/(1-\alpha)}$ is the growth-adjusted complementarity coefficient that holds the wealth-effect strength fixed in de-trended units, and
3. The saving policy, which governs the evolution of the wealth distribution.

Through channel (3), a decline in $Z_{L,t}$ reduces the rate at which productivity-wealth sorting accumulates, altering the dynamics of wealth inequality. This distinguishes $Z_{L,t}$ from a [Chari et al. \(2007\)](#) labor wedge, which operates only through the intratemporal margin and has no distributional content.

Identification through effective labor input $Z_{L,t}$ enters the production function only through $L_t \equiv Z_{L,t} \int z n(x; X) d\Phi$ (effective labor input), not as a separate factor multiplying $L^{1-\alpha}$. This rules out the labor-augmenting interpretation: a shift in $Z_{L,t}$ rescales effective hours one-for-one without changing the marginal-product schedule of L given L . In the inversion (main text Section 4), the observed labor series is interpreted as the model’s effective labor aggregate L_t , so that A_t is recovered period by period from $A_t = Y_t / (K_{eff,t}^\alpha L_t^{1-\alpha})$. $Z_{L,t}$ is then identified residually from the household labor-supply identity given $(A_t, \beta_t, K_t, \text{cov}_t(z, a), \text{cov}_t(e, a))$. The asymmetry is testable: a shift in $Z_{L,t}$ alters hours and investment composition without altering technology.

External validation Section 4.5 reports correlations between $Z_{L,t}$ and labor market indicators not used in the inversion: the employment-population ratio (level $\rho = 0.48$, HP-cycle $\rho = 0.81$), the labor force participation rate (0.28, 0.54), and average weekly manufacturing hours (0.42, 0.41). Only GDP, the capital stock, and per-capita hours enter the inversion. The strong HP-cycle correlations confirm that $Z_{L,t}$ tracks the extensive margin of labor utilization at business-cycle frequencies—exactly the margin it is designed to capture—and the moderate level correlations align $Z_{L,t}$ with secular movements in employment intensity, supporting its interpretation as a structural utilization-and-matching wedge.

Equally informative is the *negative* HP-cycle correlation ($\rho = -0.70$) with the BLS labor composition index (MPU4900092, 1987–2019), which measures workforce quality through demographics. In recessions, low-productivity workers are disproportionately separated, raising average quality—the “cleansing” effect of [Caballero and Hammour \(1994\)](#). Meanwhile, $Z_{L,t}$ declines through underutilization, matching deterioration, and foregone on-the-job learning. The sign reversal confirms that $Z_{L,t}$ captures a *utilization* margin distinct from demographic composition.

Comparison with the CKM labor wedge In [Chari et al. \(2007\)](#), the labor wedge $1 - \tau_L$ measures the gap between the intratemporal optimality condition and data but does not identify its source. In contrast, $Z_{L,t}$ has a specific structural role: it is the mean of the idiosyncratic productivity distribution, and its variation maps to composition, matching, and utilization channels measurable independently. The external validation above confirms

this structural content is empirically relevant, not merely a relabeling of the CKM residual.

E.3 Aggregate dynamics

This appendix provides the full details of the aggregate dynamics results summarized in the main text (Section 5.1.1).

Investment decomposition Prediction **P1** (approximate aggregation) can be tested directly. Investment decomposes as:

$$I^{\text{rep}} = \beta(1 + \tilde{r} + \tilde{w}Z_L\xi)K + T - (1 - \delta_t)K \quad (\text{representative-agent channel}) \quad (18)$$

$$I^{\text{ret}} = \beta\tilde{r} \text{cov}(e, a) \quad (\text{return-sorting channel}) \quad (19)$$

$$I^{\text{prod}} = \beta\tilde{w}Z_L\xi \text{cov}(z, a) \quad (\text{productivity-sorting channel}) \quad (20)$$

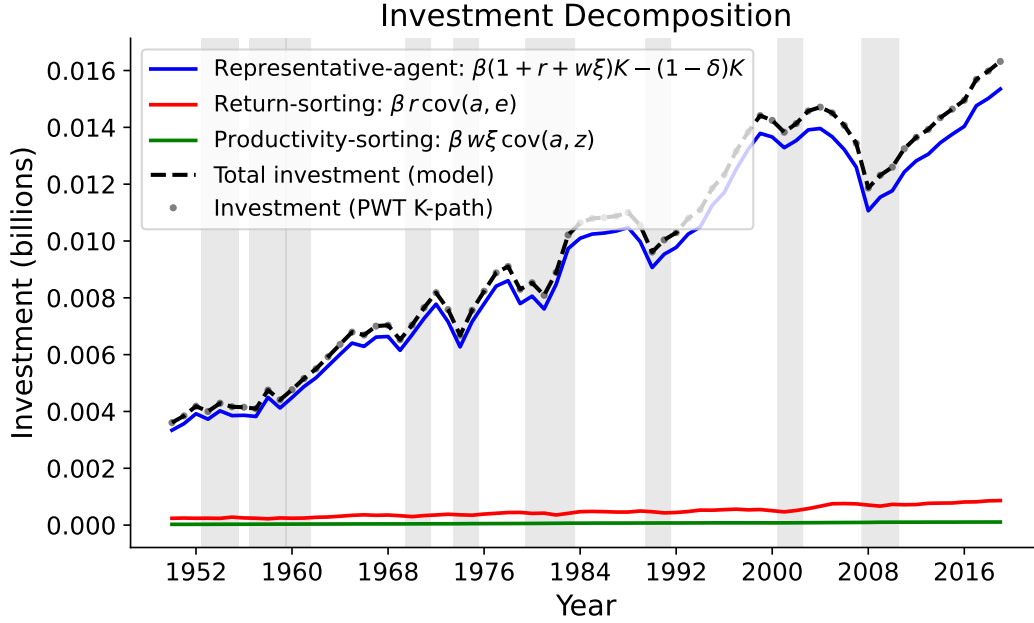
Figure E.2 plots the three channels. The representative-agent channel accounts for 93.2 percent of mean investment, confirming analytically what Krusell and Smith (1998) found numerically. The distributional channels contribute the remaining share, yet they generate the state-dependent amplification that a representative-agent model cannot produce.

Note that model-implied investment $I_t^{\text{model}} = K_{t+1} - (1 - \delta_t)K_t$ need not coincide with measured gross investment from the national accounts. The model abstracts from inventory investment, government investment, and residential construction included in NIPA aggregates.

State-dependent dynamics To test prediction **P2** (state-dependent amplification), we compute generalized impulse response functions (GIRFs) for a 2 percent negative TFP shock launched from two initial states: the 1952 distribution (low inequality, $\text{cov}(e, a) = 0.001$) and the 2019 distribution (high inequality, $\text{cov}(e, a) = 0.006$). Following Koop et al. (1996), each GIRF averages over 1,000 stochastic TFP paths with common innovations. The impact effect on output is 3 percent larger from 2019 (-2.29% vs. -2.22%), and the cumulative 10-year loss is modestly larger (-20.4% vs. -20.2%). The amplification is quantitatively small at the baseline calibration ($\sigma_e = 0.01$), consistent with the negligible return risk premium.

The mechanism operates through three reinforcing channels. First, the *exposure channel*: the return-sorting term $\beta\tilde{r} \text{cov}(e, a)$ in the capital recursion means that aggregate investment depends on the covariance between returns and wealth. When $\text{cov}(e, a)$ is large, a larger share of saving originates from households most sensitive to the aggregate return $\tilde{r}$, amplifying the investment contraction.

Figure E.2: Decomposition of aggregate investment into three channels



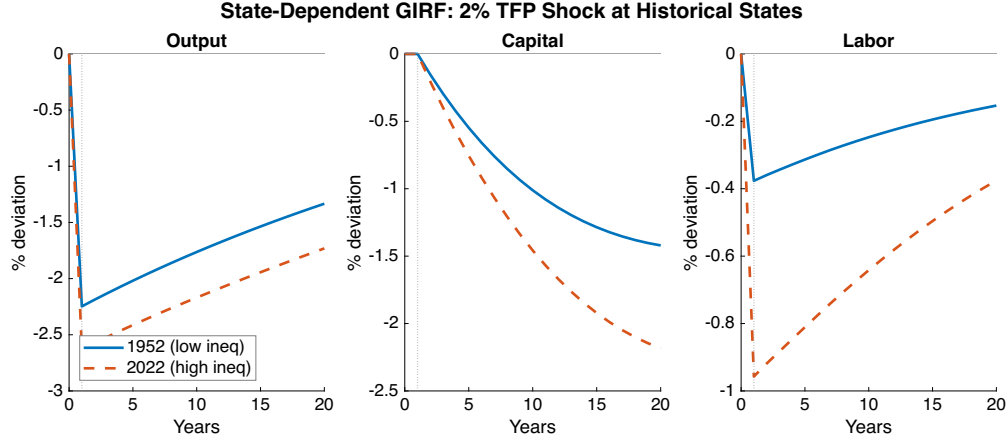
Notes: The three channels are the representative-agent channel (I^{rep}), the return-sorting channel (I^{ret}), and the productivity-sorting channel (I^{prod}). The representative-agent channel accounts for 93.2% of mean investment. Gray dots show measured NIPA investment for reference; the gap reflects model abstractions (inventories, government and residential investment). Shaded areas indicate NBER recession dates.

Second, the *labor withdrawal channel*: the wealth-effect term $-\frac{\theta(1-\beta)}{(1+\theta)\bar{w}Z_L}\tilde{r}\text{cov}(e, a)$ shows that return-wealth sorting reduces aggregate labor supply. When $\text{cov}(e, a)$ is large, wealthy households withdraw more labor through wealth-leisure complementarity ($\kappa + \xi a$), depressing output beyond the direct capital channel.

Third, the *persistence channel*: higher capital raises returns, which feeds back into future return-wealth sorting through the individual saving policy. The distributional dynamics exhibit near-unit-root persistence along the historical path, so the distributional state does not mean-revert. Sorting that amplified the initial shock *persists*, keeping the economy on a lower trajectory. At the baseline calibration, the gap between high- and low-inequality GIRFs is small (1 percent in cumulative losses), reflecting that $\sigma_e = 0.01$ generates only modest return-wealth sorting. Greater return dispersion would amplify state dependence substantially. Figure E.3 displays the GIRFs.

Return risk premium The interest rate effect of return-wealth sorting is -0.001 basis points at $\sigma_e = 0.01$ —negligible. The premium grows quadratically with σ_e but requires $\sigma_e \approx 0.83$ to reach 1 basis point. This identifies the amplification channel, not the pricing channel, as the economically significant consequence of return heterogeneity at realistic calibrations.

Figure E.3: State-dependent GIRFs to a 2% negative TFP shock



Notes: Averaged over 1,000 stochastic TFP paths. Solid: launched from the 1952 distributional state (low inequality). Dashed: launched from the 2019 distributional state (high inequality). The dotted vertical line marks the shock arrival.

The full sensitivity analysis appears in Appendix C.3.

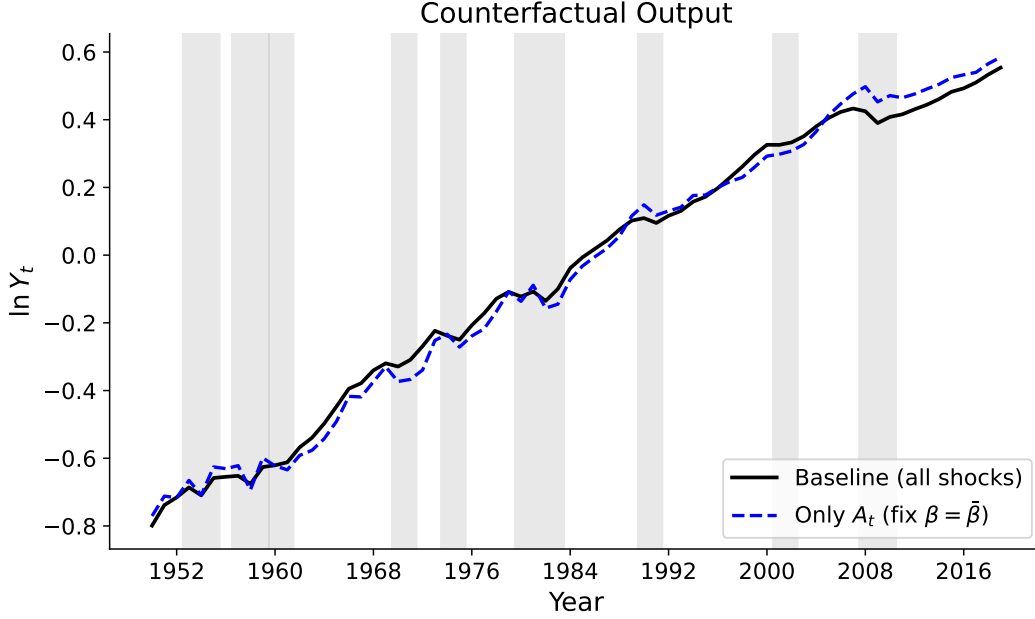
Counterfactual analysis With three structural shocks, I compare the baseline (all historical shocks) with three counterfactual economies—fixing β_t , fixing $Z_{L,t}$, or fixing both—to isolate each shock’s marginal contribution. Figure E.4 compares log output paths. Three findings stand out. First, fixing $Z_{L,t}$ at its sample mean ($\bar{Z}_L = 0.966$) yields a covariance share of +55 percent. Second, fixing β_t at its sample mean ($\bar{\beta} = 0.967$) yields −24 percent: patience partially offsets output cycles. Third, fixing both (TFP only) yields a standard-deviation ratio of 0.68 against the baseline. This confirms that aggregate TFP is the primary driver of business-cycle amplitude—consistent with [Krusell and Smith \(1998\)](#)—while $Z_{L,t}$ and β_t contribute meaningful but secondary variation.

TFP is the dominant driver of wages and the long-run interest rate, $Z_{L,t}$ is the primary driver of hours fluctuations, and β_t partially offsets TFP-driven investment cycles through countercyclical patience.

E.4 Counterfactual multi-variable panels

Figure E.5 extends the counterfactual comparison to output, labor supply, the interest rate, and the wage.

Figure E.4: Counterfactual output paths



Notes: Baseline: all structural shocks at historical values. Fix β : β_t fixed at sample mean $\bar{\beta} = 0.967$, only $(A_t, Z_{L,t})$ vary. Fix Z_L : $Z_{L,t}$ fixed at sample mean $\bar{Z}_L = 0.966$, only (A_t, β_t) vary. Fix both: only A_t varies. Shaded areas indicate NBER recession dates.

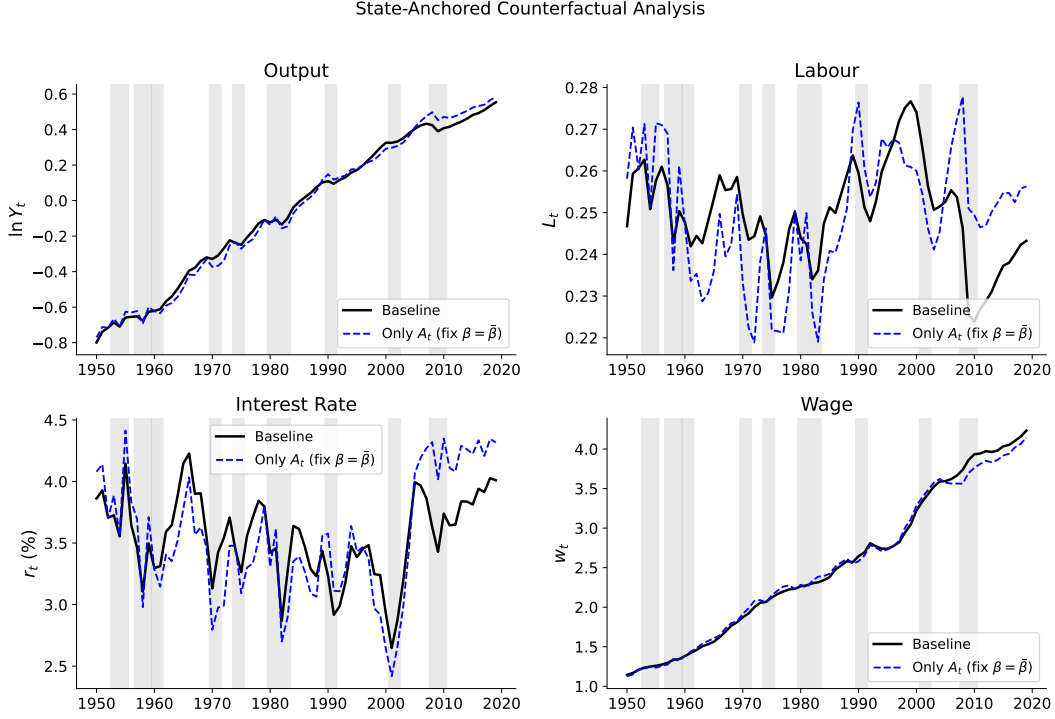
E.5 Transition dynamics

Figure E.6 plots the dominant eigenvalue governing distributional persistence over time. The dominant eigenvalue has mean 1.02 and exceeds unity in 100 percent of sample years, underscoring the extreme persistence of distributional dynamics. The near-unit-root behavior generates endogenous persistence in aggregate dynamics that converts transitory shocks into persistent output movements.

E.6 Distributional state dynamics

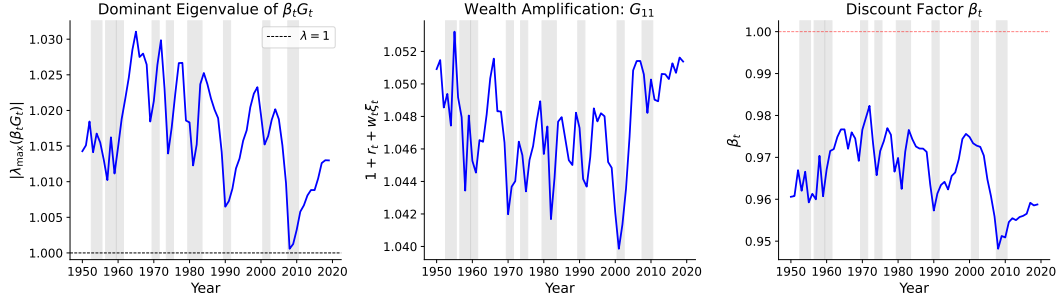
Figure E.7 plots the two covariances and their capital-normalized counterparts. From the moment-recursion inversion, the productivity-wealth covariance $\text{cov}_t(z, a)$ increases from 0.032 in 1950 to 0.133 in 2019 (a factor of 4.2), while $\text{cov}_t(e, a)$ grows from 0.256 to 0.891 (a factor of 3.5). Both are procyclical. The normalized ratio $\text{cov}(e, a)/K$ remains nearly constant, reflecting the near-unit-root dynamics of e .

Figure E.5: Counterfactual comparison across macroeconomic variables



Notes: Panels show output (log), labor supply, interest rate, and wage. Each panel compares the baseline with the fix- β counterfactual. Shaded areas indicate NBER recession dates.

Figure E.6: Transition dynamics



Notes: Top: dominant eigenvalue governing distributional persistence. Middle: wealth amplification factor. Bottom: time-varying discount factor. Shaded areas indicate NBER recession dates.

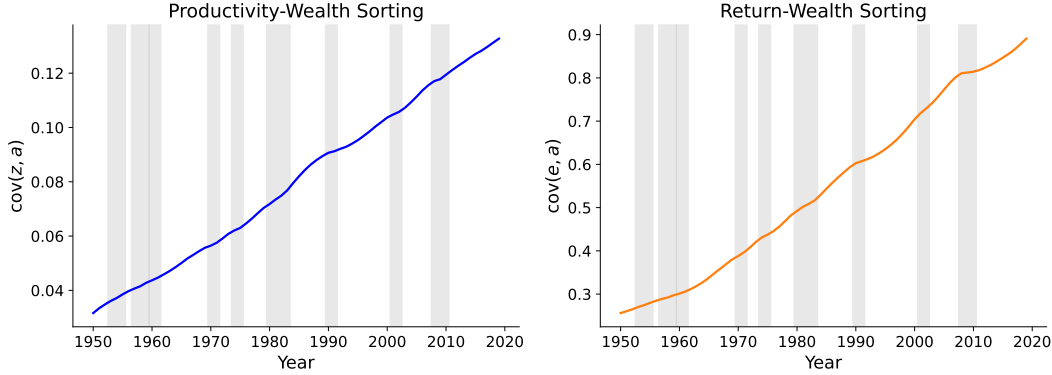
E.7 Consumption dynamics

The MRCE delivers closed-form aggregate consumption:

$$C_t = \frac{1 - \beta_t}{1 + \theta} \left[(1 + \tilde{r}_t + \tilde{w}_t Z_{L,t} \tilde{\xi}_t) K_t + \tilde{r}_t \text{cov}_t(e, a) + \tilde{w}_t Z_{L,t} \tilde{\xi}_t \text{cov}_t(z, a) \right] + \kappa \tilde{w}_t Z_{L,t},$$

where $\tilde{\xi}_t \equiv \xi / A_t^{1/(1-\alpha)}$ is the growth-adjusted complementarity coefficient.

Figure E.7: Distributional state dynamics, 1950–2019



Notes: Top row: raw covariances. Bottom row: capital-normalized ratios. Shaded areas indicate NBER recession dates.

Constant marginal propensity to consume The household MPC out of a one-time transfer is $(1 - \beta_t)/(1 + \theta)$, independent of wealth. At baseline ($\beta = 0.96$, $\theta = 4$), the annual MPC is 0.8 percent; under the sequential inversion, mean $\beta_t = 0.967$ yields an average MPC of 0.665 percent. The constant MPC follows directly from linear policy functions, contrasting with models where borrowing-constrained households exhibit MPCs of 0.3–0.5 (Kaplan et al., 2018).

Procyclical MPC The time-varying MPC $(1 - \beta_t)/(1 + \theta)$ is procyclical: when β_t is low (less patience), the MPC is higher. Under the sequential inversion, mean $\beta_t = 0.967$ (std 0.0079) yields an average MPC of 0.665 percent.

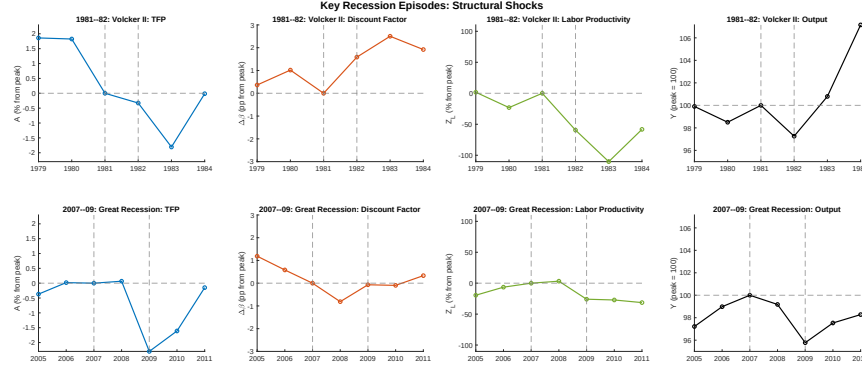
E.8 Historical episode analysis

The three-shock inversion provides a structural narrative for each recession. At annual frequency, three mild recessions (1960–61, 1969–70, 2001) produce negligible output declines and are omitted. The full classification appears in the main text (Section 5.1.1, Table 4). Figure E.8 shows shock paths for two contrasting episodes.

E.9 Cross-sectional distributional dynamics

We simulate 500,000 agents along the historical path (1950–2019) using the three-shock inversion results. Each agent i carries state (a_{it}, z_{it}, e_{it}) , where z and e follow AR(1) processes with $\sigma_z = 0.12$, $\sigma_e = 0.01$ at baseline. Countercyclical dispersion amplifies these: $\sigma_{z,t} = 1.75 \times \sigma_z$ when $Z_{L,t}$ declines (Storesletten et al., 2004), and $\sigma_{e,t} = 4.1 \times \sigma_e$ when $\Delta Y_t < 0$.

Figure E.8: Structural recession classification: shock paths around two contrasting episodes



Notes: 1981–82 (Volcker II, TFP/Labor) and 2007–09 (Great Recession, TFP-driven). Each row shows one episode. Panels show TFP (A_t , log change from peak), discount factor (β_t , pp from peak), labor productivity mean ($Z_{L,t}$, log deviation from peak), and indexed output (Y_t , peak = 100). See the main text (Table 4) for the full classification.

(Bloom et al., 2018). Wealth evolves as $a_{it+1} = \beta_t R_{it} a_{it} + T_t$ where $R_{it} = 1 + \tilde{r}_t e_{it} + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_{it}$, and consumption is $c_{it} = \frac{1-\beta_t}{1+\theta} R_{it} a_{it} + \kappa \tilde{w}_t Z_{L,t} z_{it}$.

The simulation begins after a 2,000-period burn-in at steady-state prices with GE feed-back (recomputing r , w from $K_t = \mathbb{E}[a_{it}]$ each period) and wealth truncation at $\bar{a} = 200$. Return dispersion σ_e ramps gradually from 30% to 100% of its calibrated value during the burn-in, reflecting the postwar era’s compressed return heterogeneity. Post-burn-in wealth Gini is approximately 0.76. During the historical simulation, the truncation boundary scales with capital ($\bar{a}_t = (\bar{a}/K_{ss}) \cdot K_t$) to maintain a constant max-to-mean ratio.

Distributional moments by recession appear in the main text (Table I.6). Two patterns emerge: (i) consumption inequality rises modestly in recessions (CV change +3.49%) while income CV falls (−5.42%); (ii) the Great Recession produces the highest consumption dispersion (CV 1.28, +26.5% above normal), reflecting falling β_t interacting with high wealth concentration.

E.10 Robustness

Table E.1 reports sensitivity to alternative calibrations. Approximate aggregation is extremely robust: I^{rep} accounts for 93–99 percent of total investment across all configurations. Near-unit-root persistence is robust to all parameters except the transfer share η .

Table E.1: Sensitivity of steady-state properties to alternative calibrations

Parameter	Value	K/Y	r (%)	$\lambda_{\max}$	I^{rep} (%)	Half-life (yr)
<i>Baseline</i>		3.38	2.77	0.9992	94.4	827
α	0.3	3.02	2.93	0.9991	94.0	749
	0.36	3.77	2.54	0.9992	94.8	905
	0.4	4.41	2.07	0.9993	95.7	1001
σ_e	0.001	3.24	3.17	0.9988	97.8	573
	0.005	3.29	3.02	0.9990	96.5	695
	0.02	3.54	2.32	0.9993	90.7	973
	0.05	3.86	1.54	0.9994	84.3	1157
	0.1	4.13	0.98	0.9995	79.9	1276
θ	1	3.38	2.77	0.9992	94.4	827
	1.5	3.38	2.77	0.9992	94.4	827
	3	3.38	2.77	0.9992	94.4	827
	5	3.38	2.77	0.9992	94.4	827
η	0.005	3.35	2.84	0.9997	95.0	2251
	0.01	3.37	2.79	0.9994	94.6	1110
	0.02	3.40	2.70	0.9987	93.8	541
	0.03	3.43	2.61	0.9980	93.1	353
	0.05	3.51	2.41	0.9966	91.4	204
β	0.94	2.62	5.62	0.9991	88.6	738
	0.95	2.95	4.19	0.9991	91.5	790
	0.97	3.98	1.29	0.9992	97.0	816
	0.98	5.03	-0.44	0.9992	98.1	903
ρ_z	0.9	3.37	2.79	0.9992	94.5	837
	0.93	3.37	2.78	0.9992	94.5	834
	0.97	3.40	2.71	0.9991	93.9	808
	0.99	3.65	2.04	0.9991	88.4	794

Notes: Each row varies one parameter from the baseline calibration.

F Micro inversion: data construction

This appendix documents the data construction underlying the household inversion of Section 4.3 and the household-level analyses in Sections 5.2.1–5.3. Two surveys are used. PSID 2005–2019 (8 biennial waves) supplies the panel structure required by the saving-equation identification. SCF 1992–2019 (10 triennial waves) supplies the cross-sectional intratemporal-FOC identification; the 1989 wave is excluded as a methodology outlier ($n=1,067$ vs. 11,000–18,000 in subsequent waves; $\rho(\beta_i, \log a_i)$ sign-flips against all later waves due to first-survey weighting that the AHEAD redesign fixed in 1992). All household-level variables are deflated to 2021 dollars using the CPI-U annual average and converted to model units by dividing

by $\text{dpm} \equiv 34,862 \times (\text{CPI}_{2021}/\text{CPI}_{2017}) \approx 38,545$. The factor matches the macro inversion's per-capita 2017-dollar normalization.

F.1 Common conventions

Time endowment. The labor share n_i is computed against an annual discretionary endowment of $17 \text{ h/day} \times 365 \text{ d/yr} = 6,205$ hours per worker. Households with multiple earners use the per-earner endowment scaled by the number of earners. The choice of 17 hours per day matches the time-use literature's residual after sleep and personal care, and matches the calibration in the macro side (κ, ξ) .

Macro GE inputs. For each survey wave, we read $(A_t, \beta_t^{\text{macro}}, Z_{L,t}, r_t, w_t, K_t, L_t)$ from the macro inversion's annual output (main text Section 4). After-tax prices are $\tilde{r}_t = r_t(1 - \tau_K)$ and $\tilde{w}_t = w_t(1 - \tau_L)$ at the calibrated $\tau_K = \tau_L = 0.30$. The transfer is $T_t = \eta \tau Y_t$ with $\eta = 0.0133$. The growth-adjusted complementarity is $\tilde{\xi}_t = \xi/A_t^{1/(1-\alpha)}$ with $\xi = 0.0213$, where α is the Penn World Tables capital share for year t .

Renormalization of z_i . Within each wave, we renormalize z_i so that $E_w[z_i \mid \text{worker}] = 1$. The macro per-capita wage w_t averages over the total population (workers and non-workers); the survey conditions on workers. The renormalization absorbs this universe-vs-sample wage-level gap so that cross-sectional dispersion of z_i is the relevant heterogeneity object.

Renormalization of e_i . We compute $e_i^{\text{raw}} = \text{KI}_i/(r_t a_i)$ using the gross (pre-tax) macro return r_t and PSID/SCF-reported KI_i in model units. Within each wave, we renormalize so that $E_w[e_i \mid \text{KI}_i > 0] = 1$, which absorbs both the gross-vs-net-tax convention $(1 - \tau_K)$ and any sample-level reporting biases in capital-income variables. Households with $\text{KI}_i^d \leq 0$ have e_i imputed to 1, on the interpretation that PSID/SCF capital-income items (interest, dividends) miss housing equity and retirement-account returns.

F.2 PSID biennial saving-equation derivation

The PSID identification rests on a closed-form quadratic in β_i obtained by iterating the one-period saving rule through two consecutive years to match PSID's biennial wave structure. Starting from the household policy

$$a_{i,t+1} = \beta_i a_{i,t} R_{i,t} + T_t, \quad R_{i,t} \equiv 1 + \tilde{r}_t e_i + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_i,$$

and applying it again from $t + 1$ to $t + 2$ under the (FRCE) assumption that the household treats β_i and the structural parameters as time-invariant, we obtain

$$\begin{aligned} a_{i,t+2} &= \beta_i a_{i,t+1} R_{i,t+1} + T_{t+1} \\ &= \beta_i [\beta_i a_{i,t} R_{i,t} + T_t] R_{i,t+1} + T_{t+1} \\ &= \beta_i^2 a_{i,t} R_{i,t} R_{i,t+1} + \beta_i T_t R_{i,t+1} + T_{t+1}. \end{aligned}$$

Rearranged as a quadratic in β_i with given $(a_{i,t}, a_{i,t+2}, R_{i,t}, R_{i,t+1}, T_t, T_{t+1})$,

$$[a_{i,t} R_{i,t} R_{i,t+1}] \beta_i^2 + [T_t R_{i,t+1}] \beta_i + [T_{t+1} - a_{i,t+2}] = 0. \quad (21)$$

Let $\mathcal{D} \equiv T_t^2 R_{i,t+1}^2 + 4a_{i,t} R_{i,t} R_{i,t+1} (a_{i,t+2} - T_{t+1})$ denote the discriminant. $\mathcal{D} > 0$ whenever $a_{i,t+2} > T_{t+1}$ (next-wave wealth strictly exceeds the transfer floor), which holds for every household in the consistent estimation sample. The smooth positive root

$$\beta_i = \frac{-T_t R_{i,t+1} + \sqrt{\mathcal{D}}}{2a_{i,t} R_{i,t} R_{i,t+1}}$$

is the recovered preference parameter; the negative root is discarded as economically meaningless. The main-text exposition in Section 4.3 and the implementation both collapse to the constant-price form

$$a_{i,t+2} = \beta_i^2 a_{i,t} R_{i,t}^2 + T_t (1 + \beta_i R_{i,t}),$$

which assumes $R_{i,t} = R_{i,t+1}$ and $T_t = T_{t+1}$ within the wave-pair window. This matches the forecast-invariance assumption: households treat current prices and the transfer as time-invariant when forming saving decisions. Over a biennial window with prices and transfers that move only modestly, the collapsed form is the natural empirical counterpart to the household's perceived environment.

F.3 PSID variables and sample

The PSID processed panel underlying the inversion is built from `psid_family_panel.rds` and combines biennial family-level files 2005–2019 (`family_id`, `wave_year`, `pid`). The R pipeline that produced the processed CSV lives in `empirics_psid/code/02_construct_vars_and_estimates`; the Python inversion in `exact_ge_mapping/micro_inversion.py` reads the resulting CSV directly. Key variables:

- $W_{\text{nom},i,t} = \text{Wd_nom}$: pre-tax full-time-equivalent annual wage, computed as labor income divided by n_i (the household's labor share against the per-earner endowment).

- $a_{i,t} = \text{a.m}$: net worth (PSID `net_worth_total`) deflated to 2021\$ and converted to model units. $a_{i,t+2} = \text{a_prime_m}$ is the same variable two waves ahead, deflated by the next-wave CPI and converted to model units.
- $\text{KI}_{i,t} = \text{capital_income_real}$: PSID’s reported sum of interest income, dividend income, business income, and net rental income, deflated to 2021\$.
- Survey weight = `weight_reweighted`: the PSID longitudinal family weight, renormalized within the consistent estimation sample so weights sum to the original total.
- `age_head`, `race_head`, `educ_head`: demographics on the family head, used in lifecycle and conditional analyses.

The estimation sample restricts to households satisfying `in_consistent = TRUE`: positive after-tax wage, positive labor share, positive wealth in both current and next wave, valid biennial gap, and positive longitudinal weight. Sample sizes range from 3,864 (2013) to 4,372 (2005).

F.4 SCF variables and sample

SCF data are read directly from the Federal Reserve summary extracts (`rscfpYYYY.dta`) and full extracts (`pYYi6.dta`, with `p92i4.dta` as the 1992 exception). Hours data are merged from the full extract:

- `x4110` = hours per normal week,
- `x4111` = weeks per normal year,
- `x4507` = hours per week at second job.

Pre-2013 SCF full extracts use uppercase column names (`X4110`, etc.); 2013 onwards uses lowercase. The Python loader normalizes case before extraction.

SCF reports five implicates per household to handle imputation; the survey weight `wgt` accounts for this (each implicate has weight = household weight / 5). Key variables:

- $a_{i,t} = \text{networth}$ deflated to 2021\$ then divided by `dpm`.
- $W_{\text{nom},i,t} = \text{wageinc}$ divided by the household’s labor share $n_{i,t}$ (when $n_{i,t} > 0.01$).
- $\text{KI}_{i,t} = \text{intdivinc}$ (interest + dividends) + `bussefarminc` (business and farm income) + `kginc` (capital gains), all deflated to 2021\$. SCF’s broader capital-income coverage is one reason its e_i identification is tighter than PSID’s.

- Survey weight = `wgt`.
- `age`, `edcl`, `race`, `married`: demographics.

The estimation sample restricts to households with positive net worth, positive labor share, and positive weight. We further restrict to working households ($n_i > 0.01$ and $W_{\text{nom},i} > 0$) for the intratemporal-FOC identification. Non-workers' β_i would require the consumption equation directly (with $z_i = 0$), and consumption data quality in SCF is uneven; we leave non-workers for future work.

F.5 β -clipping and quality flags (SCF only)

The SCF intratemporal-FOC formula

$$\beta_i = 1 - \frac{(1 + \theta) \tilde{w}_t Z_{L,t} z_i (\kappa + \tilde{\xi}_t a_i - n_i)}{\theta R_i a_i}$$

is sensitive to whether the implied leisure quantity $\kappa + \tilde{\xi}_t a_i - n_i$ is positive. For low-wealth households whose hours n_i exceed $\kappa + \tilde{\xi}_t a_i$ (the model-implied time endowment), the bracket flips sign and β_i exceeds unity. For numerical stability we clip β_i at $[0.5, 1.5]$. The clipping affects 1–10% of SCF households per wave (mostly very-low-wealth full-time workers); the clip indicator is reported as `beta_clipped` in the per-household output for transparency. PSID does not require clipping because the saving-equation quadratic admits a smooth positive root for any $a_{i,t+2} > T_t$.

F.6 Output files and reproducibility

The Python module `exact_ge_mapping/micro_inversion.py` produces:

- `csv_output/micro_inversion_psid.csv`: per-household-per-wave (z_i, e_i, β_i, R_i) for PSID, with demographics, weight, and the imputation indicator.
- `csv_output/micro_inversion_psid_summary.csv`: per-wave aggregate moments (sample sizes, $\beta_{\text{cap}}, \beta_{\text{pop}}, \beta_{\text{med}}$, all relevant correlations, $\widehat{\text{cov}}_t(z, a)$ and $\widehat{\text{cov}}_t(e, a)$).
- `csv_output/micro_inversion_scf.csv` and `micro_inversion_scf_summary.csv`: same structure for SCF.
- `csv_output/micro_lifecycle.csv`: 5-year-bin lifecycle profiles pooled across waves, with both surveys.

- `figures/micro_*.pdf`: six diagnostic figures (distributions by wave, scatter against $\log a$, correlation stability, PSID-vs-SCF β_{cap} , lifecycle profiles, β -macro-vs-micro time series).

The companion module `exact_ge_mapping/micro_diagnostics.py` reads the panel CSVs and produces the per-table CSV outputs used in Sections 5.2.1, 5.2.2, and 5.3:

- `micro_diag_vardecomp.csv`: variance decomposition of $\log z$ and β into person FE / age trend / wave / idiosyncratic (Table 7).
- `micro_diag_policy_invariance.csv`: pre/post wave-level moments around the four major federal tax reforms (Table 8).
- `micro_diag_delta_beta.csv`: within-person biennial $\Delta\beta_i$ statistics by wave-pair plus the pooled noise band (Section 5.2.2).
- `micro_diag_age_corr.csv`: age-bin Pearson correlations $\rho(\beta, z)$, $\rho(\beta, \log a)$, $\rho(z, \log a)$ for both surveys (Table I.8).
- `micro_quartile_table.csv`: within-survey wealth-quartile decomposition of $a, \beta, \beta_{\text{cap}}, z$ (Table I.7); unweighted a and β medians within Q4 to match the SCF dealer-sample convention.
- `micro_diag_fin_wealth.csv`: PSID inversion using `financial_wealth_real` (excluding housing equity) instead of total net worth (financial-wealth robustness footnote in Section 5.3).

The modules are invoked from `main.py` after the macro inversion completes; the macro CSV `task2e_ABZL_inversion.csv` is the only macro-side input. Re-running the macro inversion regenerates the GE inputs; re-running the micro inversion and diagnostics against an updated macro CSV requires no manual steps. PSID and SCF processed CSVs are read read-only from a sibling project directory.

G Theory: definitions, proofs, and implications

This appendix collects the formal objects and derivations underlying the equilibrium properties summarized in Section 3.3. Full proofs are located in the Online Appendix at the points indicated.

G.1 Existence and characterization of the MRCE

Definition 1 (Moment-recursive competitive equilibrium (MRCE)).

A collection $(g_c, g_a, g_n, v, \Gamma_M, K, L, w, r, \mathcal{M})$ is a moment-recursive competitive equilibrium if:

1. g_c, g_a, g_n, v solve the household's problem given Γ_M .
2. K, L satisfy the production sector's optimality conditions.
3. Factor markets clear.
4. Γ_M satisfies $\Gamma_M(\mathcal{M}(\Phi), A, Z_L, \beta) = \mathcal{M}(\Phi')$, where $\mathcal{M}$ extracts the sufficient moments from Φ , and Φ' is the distribution of individual states induced by g_a and the exogenous processes.

The key departure from the RCE is that Γ_M maps a *finite* moment vector to contemporaneous equilibrium allocations and prices. The sufficient statistics are aggregate capital K and two covariances: $\text{cov}(z, a)$ measures the alignment between earnings ability and wealth; $\text{cov}(e, a)$ measures the alignment between portfolio returns and wealth.

Proposition 2 (Unique existence of the MRCE).

The MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, \text{cov}(z, a), \text{cov}(e, a)]$. Define after-tax prices $\tilde{r} = r(\mathcal{M}, A)(1 - \tau)$, $\tilde{w} = w(\mathcal{M}, A)(1 - \tau)$, and the transfer $T = \eta \cdot Y \cdot \tau$. Then: (i) *Contemporaneous sufficiency.* Given $\mathcal{M}$ and the aggregate state (A, Z_L, β) , all equilibrium allocations (individual consumption $c(x; X)$, saving $a'(x; X)$, labor supply $n(x; X)$, and aggregate prices r, w, L) are fully determined without reference to Φ beyond its moments $\mathcal{M}$. (ii) *Aggregate capital law of motion.* K' has the exact closed form (17). (iii) *Factor prices and aggregate labor.* With $K_{eff} = K + \text{cov}(e, a)$, the factor prices r, w and aggregate labor L are as displayed in Section 3.3, equation (18).

Proof. See Appendix A.3 of the Online Appendix. ■

The K' equation reveals three distinct channels. The first term $(1 + \tilde{r} + \tilde{w}Z_L\xi)K$ is the representative-agent channel. The second, $\tilde{r} \text{cov}(e, a)$, is the *return-sorting channel*: high-return households are disproportionately wealthy, raising aggregate saving above what average returns and average wealth would predict. The third, $\tilde{w}Z_L\xi \text{cov}(z, a)$, is the *productivity-sorting channel*: productive-wealthy households generate additional saving through wealth-leisure complementarity. These terms make dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

G.2 Forecast-invariant equilibrium

Definition 2 (Forecast-invariant recursive competitive equilibrium (FRCE)).

A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, K, L, w, r)$ is a forecast-invariant recursive competitive equilibrium if:

1. g_c, g_a, g_n, v solve the household's problem given arbitrary beliefs about aggregate Markov laws of motion for both endogenous and exogenous states.
2. K, L satisfy the production sector's optimality conditions.
3. Factor markets clear.
4. Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the policy function g_a and the exogenous idiosyncratic process to Φ .

Proposition 3 (Unique existence of the FRCE).

The FRCE uniquely exists and the allocations are identical to the MRCE.

Proof. See Appendix A.7 of the Online Appendix. ■

G.3 Closure of the moment recursion

While $\mathcal{M}$ is sufficient for all contemporaneous equilibrium allocations and K' is exactly determined, the forward evolution of $\text{cov}(z', a')$ and $\text{cov}(e', a')$ generally depends on higher-order moments of the joint distribution Φ . Specifically, from $a' = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L z \xi) + T$ and $z' = \rho_z z + \varepsilon_z$:

$$\text{cov}(z', a') = \rho_z \beta [(1 + \tilde{r} + \tilde{w}Z_L \xi) \text{cov}(z, a) + \tilde{w}Z_L \xi \mathbb{E}[\hat{z}^2 a] + \tilde{r} \mathbb{E}[\hat{z} \hat{e} a]], \quad (22)$$

where $\hat{z} = z - 1$ and $\hat{e} = e - 1$. The terms $\mathbb{E}[\hat{z}^2 a]$ and $\mathbb{E}[\hat{z} \hat{e} a]$ involve three-way cross-moments that are not functions of $\mathcal{M}$ alone. Under joint normality of $(\hat{z}, \hat{e}, a)$, these reduce to $\mathbb{E}[\hat{z}^2 a] = \sigma_z^2 K$ and $\mathbb{E}[\hat{z} \hat{e} a] = 0$ (by the Isserlis/Wick theorem, using independence of z and e), and the moment dynamics close. However, the Kesten-process wealth dynamics generate Pareto tails that violate normality. Tracking the full distribution via agent-based simulation is therefore necessary for the forward evolution of distributional moments, while the exact household policies, the K' equation (17), and all contemporaneous allocations are preserved analytically. The quantitative analysis in Section 4 employs this approach: exact policies govern each agent's behavior, and the distributional moments are computed from the simulated cross-section.

G.4 Ergodicity

Corollary 1 (Kesten stationarity).

The individual law of motion $a' = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L z \xi) + T$ is a Kesten process with multiplicative coefficient $R_i = \beta((1 + \tilde{r}e) + \tilde{w}Z_L z \xi)$ and additive intercept $T > 0$. When $\mathbb{E}[\log R_i] < 0$, the wealth distribution admits a unique stationary solution with Pareto tails (Kesten, 1973). The additive transfer $T > 0$ is essential: without it, the purely multiplicative system admits only the degenerate distribution at zero.

Proof. See Appendix A.4 of the Online Appendix. ■

The near-unit-root property of the dominant eigenvalue (empirically ≈ 1.01 in the calibrated model) generates slow mean reversion of the wealth distribution and amplifies shock propagation, as shown in Section 4.

G.5 Special cases

The MRCE nests several benchmark models, clarifying what each source of heterogeneity contributes to aggregate dynamics.

Corollary 2 (Krusell and Smith (1998) analogue).

Suppose return heterogeneity is muted: $e = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, \text{cov}(z, a)]$, and the aggregate capital law of motion is:

$$K' = \beta [(1 + \tilde{r} + \tilde{w}Z_L \xi)K + \tilde{w}Z_L \xi \text{cov}(z, a)] + T \quad (23)$$

Proof. See Appendix A.5 of the Online Appendix. ■

Without return heterogeneity, the state reduces to $[K, \text{cov}(z, a)]$, the analytical counterpart of Krusell and Smith (1998). Capital alone suffices if and only if $\text{cov}(z, a) = 0$. Approximate aggregation holds when $\text{cov}(z, a)$ is small relative to K , i.e., when productivity shocks are weakly persistent or wealth mobility is high. The MRCE thus provides exact conditions under which the Krusell and Smith (1998) finding holds as a theorem. Further discussion appears in Appendix B.3.

Corollary 3 (Representative-agent analogue).

Suppose all heterogeneity is muted: $e = 1$ and $z = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = K$, and the law of motion is:

$$K' = \beta [(1 + \tilde{r} + \tilde{w}Z_L \xi)K] + T, \quad (24)$$

which leads to the aggregate consumption function:

$$C(\mathcal{M}, A, Z_L, \beta) = \frac{1 - \beta}{1 + \theta} (1 + \tilde{r} + \tilde{w} Z_L \xi) K + \kappa \tilde{w} Z_L. \quad (25)$$

Proof. See Appendix A.5 of the Online Appendix. ■

With all heterogeneity eliminated, both covariances vanish and the model reduces to the stochastic growth model with log preferences and endogenous labor supply (Brock and Mirman, 1972). Every departure of the full MRCE from this benchmark is therefore attributable to sorting between wealth and idiosyncratic characteristics.

G.6 Macroeconomic implications

State-dependent aggregate dynamics The capital law of motion (17) decomposes into three channels: the representative-agent channel $(1 + \tilde{r} + \tilde{w} Z_L \xi) K$, the return-sorting channel $\tilde{r} \text{cov}(e, a)$, and the productivity-sorting channel $\tilde{w} Z_L \xi \text{cov}(z, a)$. These distributional terms make aggregate dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional channels The aggregate consumption function decomposes as:

$$C(\mathcal{M}, A, Z_L, \beta) = \underbrace{\frac{1 - \beta}{1 + \theta} (1 + \tilde{r} + \tilde{w} Z_L \xi) K + \kappa \tilde{w} Z_L}_{\text{Representative-agent level}} + \underbrace{\frac{1 - \beta}{1 + \theta} (\tilde{r} \text{cov}(e, a) + \tilde{w} Z_L \xi \text{cov}(z, a))}_{\text{Distributional channel}} \quad (26)$$

The aggregate employment equation reveals a parallel decomposition:

$$L(\mathcal{M}, A, Z_L, \beta) = Z_L \kappa + \underbrace{\frac{Z_L \xi (1 + \theta \beta) (K + \text{cov}(z, a))}{1 + \theta}}_{\text{Leisure enhancement}} - \underbrace{\frac{\theta (1 - \beta) [K (1 + \tilde{r}) + \tilde{r} \overbrace{\text{cov}(e, a)}^{\text{Dist. channel}}]}{(1 + \theta) \tilde{w}}}_{\text{Wealth effect}} \quad (27)$$

The wealth-effect term is amplified by $\text{cov}(e, a)$: wealthy households earning high returns consume more leisure, reducing labor supply. In periods of high return-wealth sorting, a positive productivity shock elicits a smaller employment response as the households with the largest wealth gains withdraw the most labor.

State-dependent inequality dynamics The K' equation (17) generates a “wealth spiral”: productivity shocks raise capital, which strengthens return-wealth sorting through $\text{cov}(e, a)$, which raises future capital. Discounting ($\beta < 1$) and the transfer T counterbal-

ance. The persistence of the sorting mechanism, governed by the near-unit-root dynamics of the wealth distribution, controls the speed of concentration (Appendix B.4).

G.7 Asset pricing implications

Individual return heterogeneity, traditionally viewed as diversifiable, affects aggregate prices once it is sorted with wealth (Constantinides and Duffie, 1996; Storesletten et al., 2004). Households with persistently high returns accumulate disproportionate wealth, shifting labor supply and equilibrium returns:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \text{cov}(e, a)} < 0 \quad (28)$$

Higher return-wealth sorting reduces the equilibrium interest rate through two channels: a static one (the wealth effect lowers labor supply, raising the capital-labor ratio) and a dynamic one (persistent sorting accelerates capital accumulation, depressing future returns). The full decomposition and quantitative implications appear in Appendix C.

G.8 Extension: a household-level leisure weight θ_i

The block-triangular structure of the household inversion (Section 4.4) leaves one PSID equation unused. Once z_i , e_i , and β_i are recovered from the wage, capital income, and the two-period wealth change, the intratemporal labor optimality condition is free, and it identifies a household-specific leisure weight θ_i at fixed (κ, ξ) . This turns the intensive-margin elasticity, normally imposed, into a recovered cross-sectional object.

Substituting the closed-form consumption policy of Proposition 1 into the labor condition and using the recovered (z_i, e_i, β_i) eliminates consumption, so the exercise inherits the inversion's no-consumption-data property. The $\kappa(1 + \theta)$ and $-\theta\kappa$ terms cancel, and the condition collapses to

$$\underbrace{\kappa + \tilde{\xi}_t a_i - n_i}_{\text{leisure } \ell_i} = \frac{\theta_i}{1 + \theta_i} (1 - \beta_i) \frac{a_i R_i}{\tilde{w}_t Z_{L,t} z_i}, \quad R_i = 1 + \tilde{r}_t e_i + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_i, \quad (29)$$

which is the same labor equation that yields β_i^{SCF} in (22), now solved for θ_i rather than β_i . Writing the right-hand factor as the dimensionless ratio

$$q_i \equiv \frac{(\kappa + \tilde{\xi}_t a_i - n_i) \tilde{w}_t Z_{L,t} z_i}{(1 - \beta_i) a_i R_i}, \quad (30)$$

the leisure-to-resources ratio of the household (its leisure ℓ_i relative to the wage-deflated value of wealth-financed consumption), (29) inverts in closed form:

$$\theta_i = \frac{q_i}{1 - q_i}. \quad (31)$$

The map is monotone: at a given wealth and wage, a household that takes more leisure (higher q_i) reveals a stronger taste for it. Admissibility requires $q_i \in (0, 1)$. The lower bound holds automatically for any interior, working household, since the numerator is leisure $\ell_i > 0$. The upper bound fails only for households whose leisure exceeds the time-value of their wealth, a configuration no positive leisure weight can rationalize; these are flagged and their θ_i imputed to the calibrated θ , paralleling the treatment of zero-capital-income households in Section 4.3. Because (31) takes β_i as an input, it is intrinsically a panel exercise: the saving margin pins β_i from the wealth change, and hours then pin θ_i , two PSID observables identifying two distinct preference parameters. The leisure weight, which does not enter β_i^{PSID} in (23), is now itself recovered.

H Supplementary empirical results

These tables support the validation of Section 4.5.

Table H.2: Krusell–Smith validation: inversion accuracy

	$\text{mean} A^{\text{inv}} - A^{\text{true}} /A$	$\text{CV}(\beta)$	$\sigma(\Delta \log Z_L)$	$\text{mean} Y^{\text{inv}} - Y^{\text{true}} /Y$
KS98 baseline	7.5×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + endo labor	7.3×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + Frisch + adj cost	2.5×10^{-5}	0.05%	0.18%	$< 10^{-15}$
U.S. data (1950–2019)	—	1.78%	3.42%	exact

Notes: In the simulated economies, β and Z_L are constant by construction and only aggregate TFP varies. The final row applies the same inversion to U.S. data.

I Additional results for Section 5

This appendix collects the supporting exhibits for Section 5.

Table H.3: External validation: recovered $Z_{L,t}$ and labor-market indicators

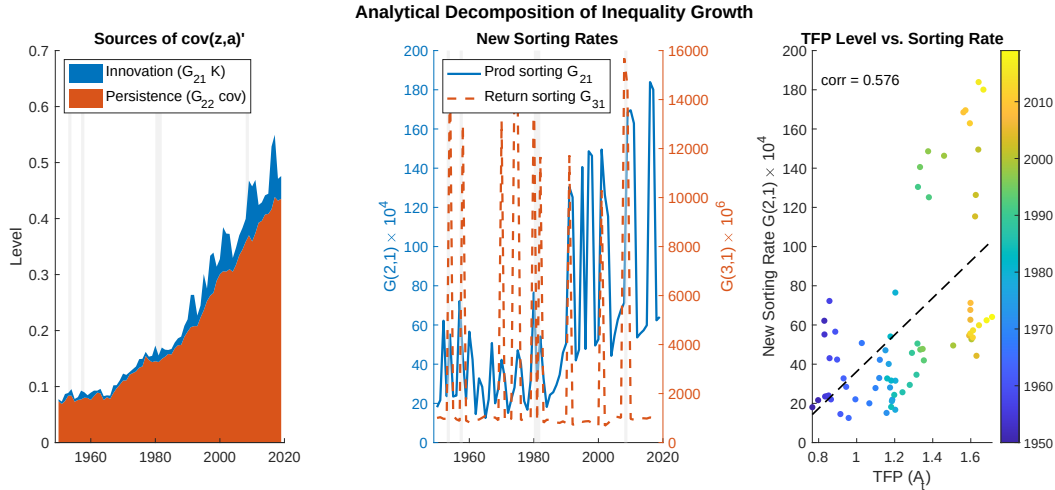
External series	Level corr	HP-cycle corr
Employment-population ratio (EMRATIO)	0.48	0.81
Labor force participation rate (CIVPART)	0.28	0.54
Avg. weekly hours, manufacturing (AWHMAN)	0.42	0.41

Notes: The external series are not used in the inversion. HP-cycle correlations use annual data with $\lambda = 6.25$.

I.1 Sorting accelerator: decomposition, incidence, and cross-sectional dynamics

Analytical decomposition The MRCE makes the sorting accelerator precise. In the return-sorting channel, persistence is governed by $\phi_e \rho_e (1 + \tilde{r})$, which is close to $1/\beta$ because return shocks are highly persistent. In the productivity-sorting channel, persistence is lower, but countercyclical dispersion raises the innovation rate during downturns. Figure I.9 decomposes these forces. The secular component comes from persistent return heterogeneity; the cyclical component comes from recession-driven increases in idiosyncratic dispersion.

Figure I.9: Analytical decomposition of distributional sorting



Notes: Left: innovation and persistence contributions to $\text{cov}(z, a)'$. Center: sorting-rate time paths for productivity and return shocks. Right: TFP level against productivity sorting. Shaded areas indicate NBER recession dates.

Who gains from TFP growth? Table I.4 reports the impact of 1 percent TFP growth by wealth percentile. The top decile captures 81.6 percent of the aggregate wealth gain, exceeding its 77.2 percent wealth share, because the return-sorting and productivity-sorting

channels compound. The top 1 percent alone captures 68.1 percent. The table is not a tax counterfactual; it is a sufficient-statistic calculation showing how the analytical channels allocate a common productivity improvement across the wealth distribution.

Table I.4: Distributional impact of 1% TFP growth at steady state

Group	Wealth	Gain	Mean Gain	Channel (%)		
	Share (%)	Share (%)		Capital	Labor	Transfer
Bottom 10%	1.2	0.8	0.0806	65.1	18.9	16.0
10–20%	1.4	1.0	0.0850	67.9	19.8	12.3
20–30%	1.7	1.2	0.0873	69.4	20.2	10.4
30–40%	1.9	1.4	0.0893	70.6	20.5	8.9
40–50%	2.2	1.7	0.0913	71.7	20.6	7.6
50–60%	2.5	2.0	0.0932	72.8	20.7	6.5
60–70%	3.0	2.4	0.0954	74.0	20.7	5.3
70–80%	3.7	3.1	0.0982	75.4	20.5	4.1
80–90%	5.3	4.6	0.1019	77.2	20.1	2.8
Top 10%	77.2	81.6	0.1102	84.0	15.9	0.2
Top 5%	72.8	77.6	0.1138	84.2	15.7	0.1
Top 1%	63.1	68.1	0.1195	84.6	15.4	0.0

Notes: Gains computed from 1% TFP growth at the stationary distribution of simulated households. “Capital” is the gain from higher after-tax capital income; “Labor” is the gain from higher labor income; “Transfer” is the common transfer component.

Historical gain shares The steady-state calculation is conservative because it holds the distribution fixed. Along the historical path, the distribution itself changes. Table I.5 reports decade averages of TFP-driven wealth gains by wealth decile, and Figure I.10 visualizes the evolving distribution. The top decile captures approximately 67.3 percent of TFP-driven wealth gains in the 1950s, rising to 69.4 percent by the 2010s. This increase reflects the endogenous accumulation of return-wealth sorting. As the wealth distribution concentrates through the Kesten mechanism, the return-sorting and productivity-sorting channels compound through the exact saving rule $a' = \beta R_i a + T$, and the top decile’s share of any marginal productivity improvement grows accordingly.

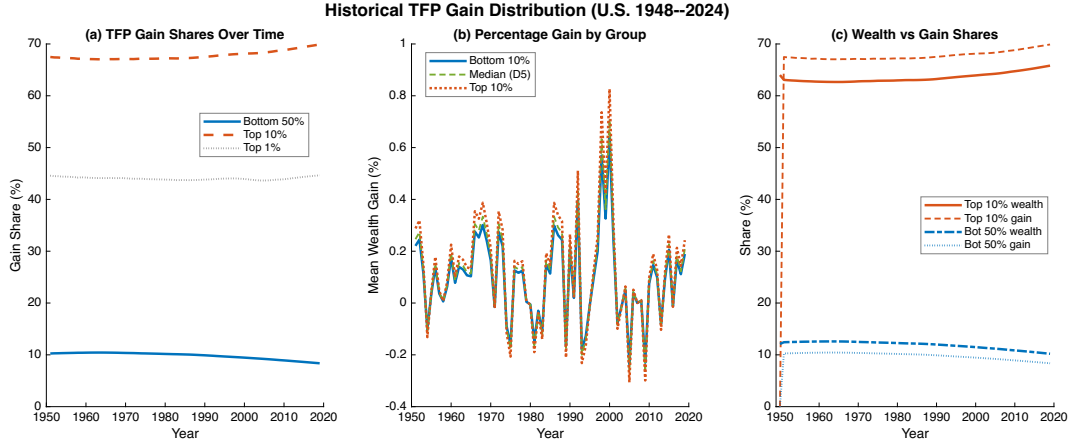
Cross-sectional inequality dynamics Figure I.11 summarizes the time-series path of wealth, consumption, and income inequality. Three patterns stand out. First, wealth concentration rises monotonically from 0.827 to 0.876. The driver is the near-unit-root persistence of return-wealth sorting, the same Kesten mechanism that generates the sorting accelerator. Second, consumption inequality also rises (from 0.281 to 0.356). The consumption rule

Table I.5: Share of TFP-driven wealth gains by wealth decile (%), decade averages

Decade	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Top 5%	Top 1%
1950s	1.3	1.7	2.1	2.4	2.9	3.5	4.4	5.7	8.8	67.3	59.7	44.4
1960s	1.3	1.7	2.1	2.5	2.9	3.5	4.4	5.8	8.8	67.1	59.4	44.1
1970s	1.2	1.7	2.0	2.4	2.9	3.5	4.4	5.8	8.9	67.1	59.4	44.0
1980s	1.2	1.6	2.0	2.4	2.9	3.5	4.4	5.8	8.9	67.3	59.5	43.8
1990s	1.1	1.6	1.9	2.3	2.8	3.4	4.3	5.8	8.9	67.8	59.9	44.0
2000s	1.0	1.5	1.8	2.2	2.7	3.3	4.3	5.8	9.0	68.4	60.3	43.8
2010s	0.9	1.3	1.7	2.1	2.6	3.2	4.1	5.7	9.0	69.4	61.2	44.3
Full sample	1.1	1.6	1.9	2.3	2.8	3.4	4.3	5.8	8.9	67.8	59.9	44.0

Notes: Each entry reports the average share of aggregate TFP-driven wealth gains captured by the indicated wealth decile. Gains are computed period by period along the historical simulation.

Figure I.10: Historical distribution of TFP-driven gains

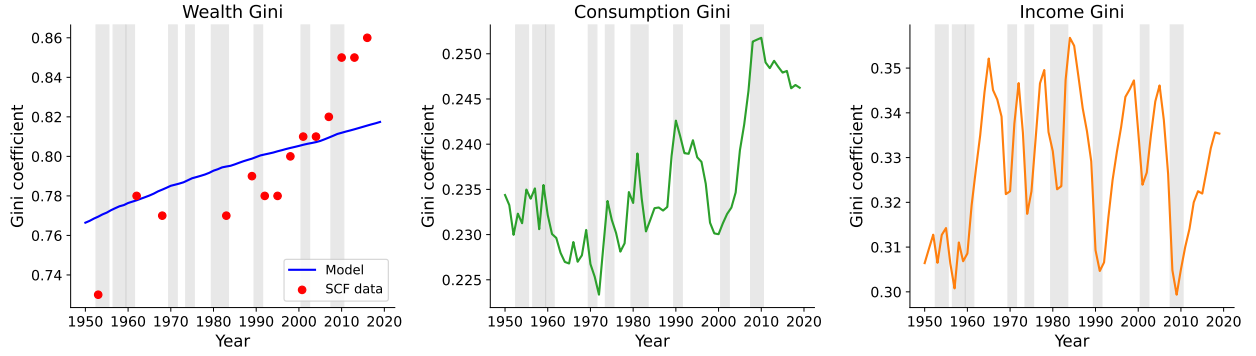


Notes: Period-by-period decomposition of the share of TFP-driven wealth gains by wealth percentile, U.S. 1950–2019.

$c_i \propto (1 - \beta)R_i a_i$ loads directly on wealth, so cross-sectional dispersion is amplified as the wealth distribution concentrates. Third, income inequality is flatter (mean Gini 0.404). Two offsetting forces are at work: rising wealth inequality pushes capital income apart, but the labor component, which scales with $Z_{L,t}z_i$ rather than wealth, compresses during recessions when low-productivity workers exit. The divergence between consumption and income Gini is a testable implication of the MRCE’s exact policy structure.

The time-series Gini compresses the distribution into one number. Figure I.12 overlays the full cross-sectional wealth distribution (normalized by mean wealth) in 1980 and 2015. The 2015 distribution exhibits a fatter right tail and a thinner middle: the mass between 0.5 and 1.5 times mean wealth declines while the upper tail extends. This is the distributional signature of the sorting accelerator: three decades of persistent return heterogeneity stretch

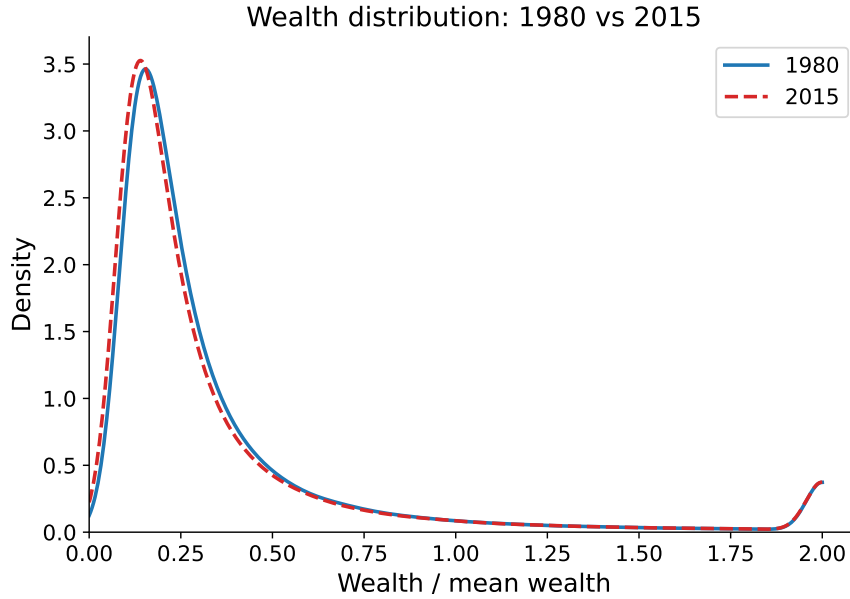
Figure I.11: Historical cross-sectional distributional dynamics



Notes: Historical simulation using the recovered aggregate shocks and exact household policies. The wealth panel includes SCF-based estimates as external benchmarks. Shaded areas indicate NBER recession dates.

the Pareto tail while the additive transfer T keeps the lower tail anchored.

Figure I.12: Normalized wealth distribution: 1980 vs. 2015



Notes: Kernel density estimates of wealth divided by mean wealth from the historical household simulation. Values are winsorized at twice mean wealth for readability.

Table I.6 reports consumption and income moments in normal times and across recession episodes. Two patterns emerge. First, consumption inequality rises in recessions (+3.49% in CV on average) while income inequality falls (−5.42%). The consumption response reflects the model’s exact policy structure: when β_t falls (rising impatience), the consumption share $(1 - \beta_t)/(1 + \theta)$ increases, amplifying the wealth-driven component $R_i a_i$ in cross-sectional consumption. Income compression reflects declining $Z_{L,t}$, which reduces labor-income dis-

persion by scaling down the $Z_L z$ term. Second, the magnitude of the consumption response varies sharply across episodes. The Great Recession stands out: consumption CV rises 26.5% above normal times, driven by the interaction of high wealth concentration ($\text{cov}(e, a)$ at its sample maximum) with a sharp fall in the saving-margin wedge. By contrast, the 1953–58 recessions show consumption CV *below* normal, reflecting the lower wealth concentration of the early postwar period.

Table I.6: Cross-sectional distributional moments: normal times vs. recessions

	Consumption				Income			
	CV	Gini	Skew	Kurt	CV	Gini	Skew	Kurt
Normal	1.009	0.334	8.19	92.3	1.798	0.408	11.75	178.7
1953–54 (Korean War)	1.003	0.291	10.28	134.2	1.738	0.359	12.89	203.6
1957–58 (Eisenhower)	0.973	0.303	9.42	118.1	1.600	0.363	12.30	189.7
1973–75 (Oil Crisis)	0.897	0.331	7.13	75.9	1.708	0.402	11.94	185.0
1980 (Volcker I)	0.928	0.318	8.15	92.1	1.742	0.394	11.67	167.9
1981–82 (Volcker II)	0.967	0.327	8.16	92.1	1.671	0.393	11.84	182.8
1990–91 (S&L Crisis)	1.168	0.340	9.58	114.8	1.696	0.392	11.99	187.3
2007–09 (Great Recession)	1.276	0.365	9.34	107.4	1.744	0.411	11.80	191.5

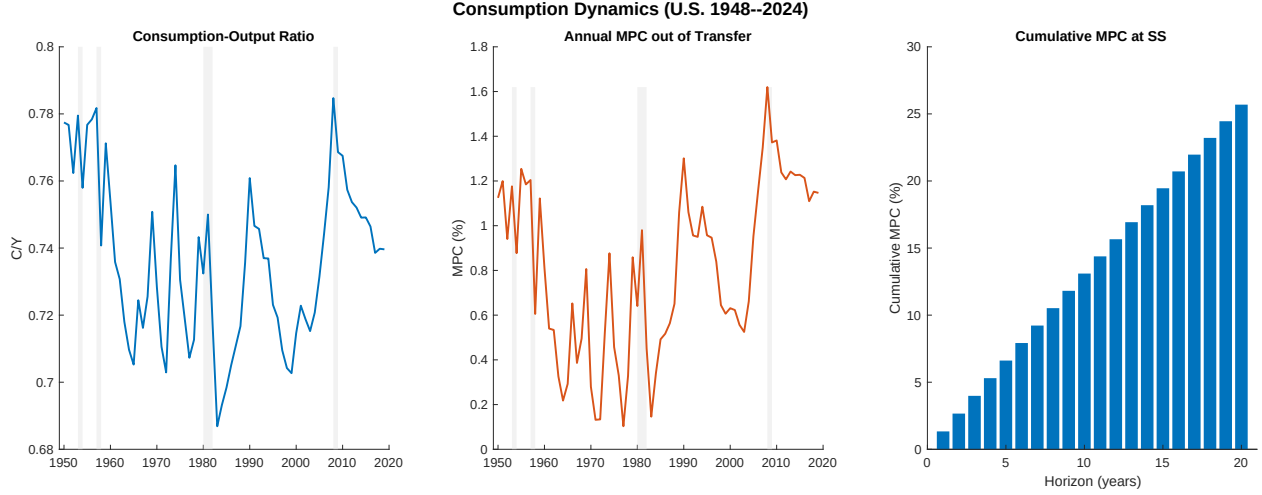
Notes: Cross-sectional moments from simulating households along the historical equilibrium path using the three-shock inversion with countercyclical dispersion. CV = coefficient of variation. Normal times are non-NBER-recession years; recession moments are averaged within each episode.

Consumption dynamics The MRCE also yields a closed-form aggregate consumption equation,

$$C_t = \frac{1 - \beta_t}{1 + \theta} \left[(1 + \tilde{r}_t + \tilde{w}_t Z_{L,t} \tilde{\xi}_t) K_t + \tilde{r}_t \text{cov}_t(e, a) + \tilde{w}_t Z_{L,t} \tilde{\xi}_t \text{cov}_t(z, a) \right] + \kappa \tilde{w}_t Z_{L,t},$$

where $\tilde{\xi}_t \equiv \xi / A_t^{1/(1-\alpha)}$ is the growth-adjusted complementarity coefficient (see Online Appendix B). The individual MPC out of a one-time transfer is $(1 - \beta_t)/(1 + \theta)$, independent of wealth; it is a constant that varies only with the aggregate saving-margin wedge. Figure I.13 reports three objects. The left panel shows the consumption-output ratio, which rises from 0.69 in the 1980s to 0.78 by 2010, tracking the secular decline in β_t (lower patience $\rightarrow$ higher consumption share). The center panel plots the annual MPC, which mirrors β_t : the MPC spikes during the 2007–09 recession when patience falls sharply. The right panel shows the cumulative MPC at steady state reaching approximately 25% after 20 years. The low annual MPC ($\sim 0.665\%$) is a scope condition: the MRCE isolates aggregation and sorting rather than liquidity-channel consumption responses (Kaplan et al., 2018).

Figure I.13: Consumption dynamics and transfer MPC



Notes: Left: consumption-output ratio. Center: annual MPC out of a one-time transfer implied by $(1 - \beta_t)/(1 + \theta)$. Right: cumulative steady-state MPC by horizon.

I.2 Distributional impact of capital-tax increases

This figure is the tax-increase counterpart to Figure 7 in Section 5.1.4; the quantitative summary is Panel B of Table 6.

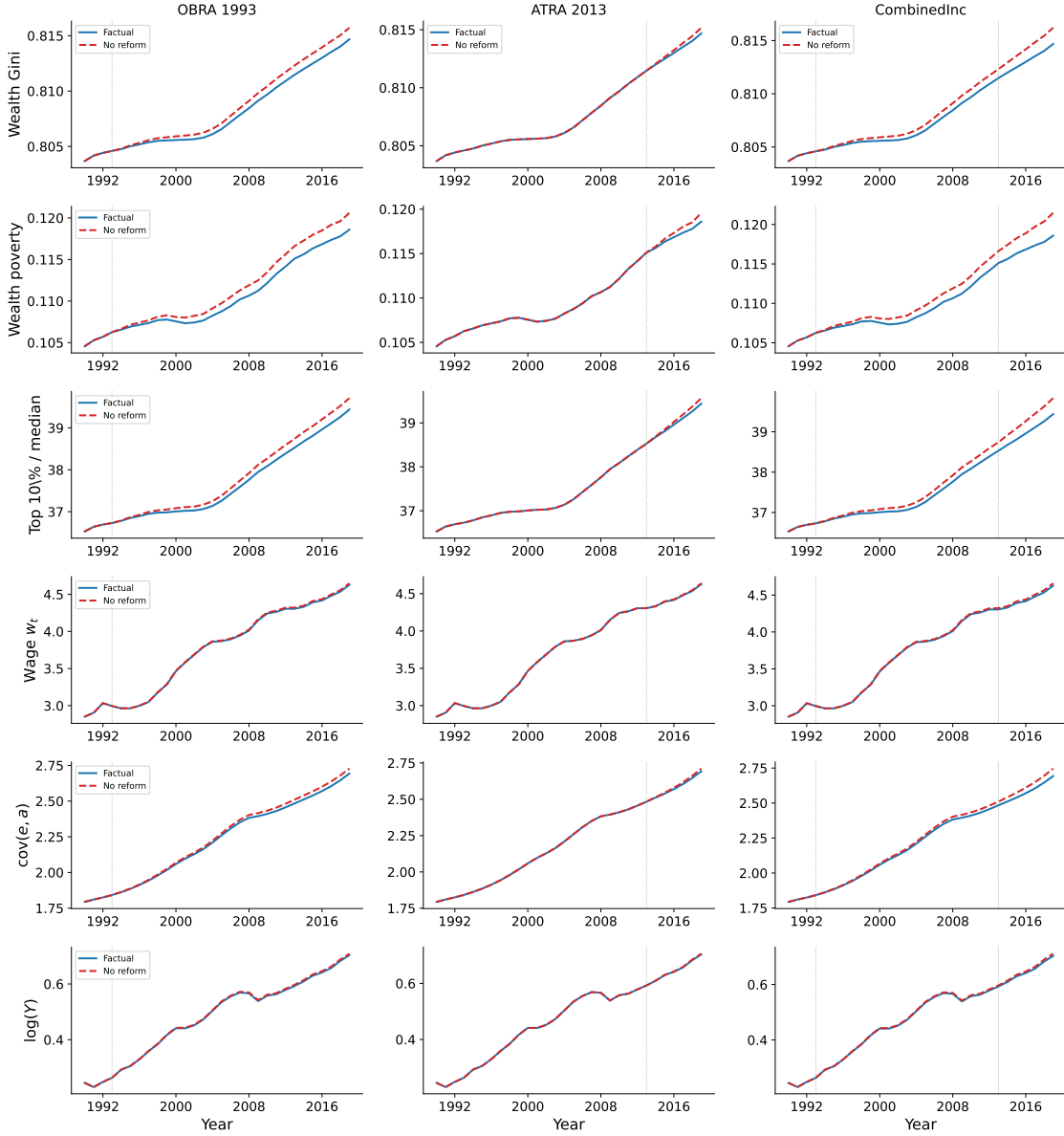
I.3 Household parameters: quartile and age-conditional detail

Distributional dynamics over three decades. The 10 SCF waves and 8 PSID waves provide direct evidence on how the cross-section of structural parameters evolves over nearly three decades. Figure I.15 plots four time series; panel (a) overlays the wave-by-wave $\hat{\beta}_{\text{cap},t}$ in PSID and SCF against the macro β_t .

Heterogeneity in β widens over time. The cross-sectional spread $P_{90}(\beta_i) - P_{10}(\beta_i)$ in SCF rises from 0.27 in 1992 to 0.46 in 2019 (Figure I.15b), a 70% increase concentrated in the post-2007 period (pre-2007 average ≈ 0.30 ; post-2007 average ≈ 0.43). PSID's spread rises in parallel, from 0.45 in 2005 to 0.61 in 2019. With both surveys on the static identification the levels are now comparable and the trend is the same in both. Household-level saving behavior has become more disparate over the same three decades that wealth concentration rose, consistent with the macro paper's sorting accelerator. (Under the dynamic route PSID's spread is mechanically much wider, ~ 1.5 , but that is identification noise, not heterogeneity; Appendix I.3.)

The wealth gradient of β is large and stable. $\rho(\beta_i, \log a_i)$ stays in $[-0.51, -0.63]$ in SCF post-1992 and around -0.30 in PSID across all waves (Figure I.15c). The wealthy persistently dissave while the poor accumulate at $\beta > 1.1$. Panel (d) of Figure I.15 shows the same

Figure I.14: Macroeconomic impact of capital tax increases (OBRA 1993, ATRA 2013)

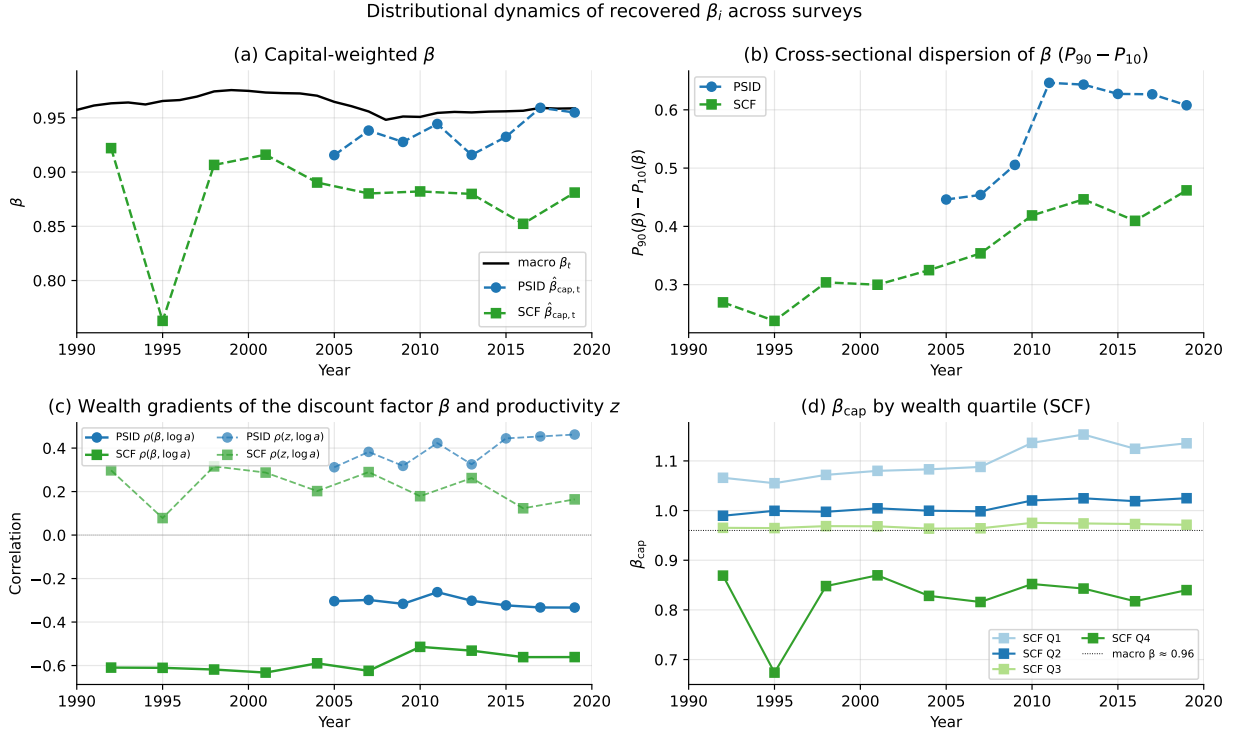


Notes: Same simulation and panel layout as Figure 7. Columns: OBRA 1993, ATRA 2013, and combined increases. Solid blue: factual; dashed red: reform-only counterfactual (subtract only the named reform's reform-year τ_K shock from the factual path going forward, keeping all subsequent reforms intact). The horizontal axis starts at 1990 to focus on the post-reform window.

gradient as $\hat{\beta}_{\text{cap}}$ by within-survey wealth quartile: it falls monotonically from the poorest to the richest quartile in every wave, a stable validation that the inversion reproduces the dis-saving of the wealthy in each cross-section. Table I.7 in Appendix I.3 reports the underlying decomposition.

PSID Q4 wealth *shrinks* at $-1.9\%/yr$, giving median $\beta = 0.98$ and capital-weighted

Figure I.15: Distributional dynamics of recovered β_i over three decades



Notes: (a) Capital-weighted $\hat{\beta}_{cap,t}$ in PSID (biennial 2005–2019) and SCF (triennial 1992–2019), with the macro β_t overlaid. Panels (a) and (d) report $\hat{\beta}_{cap}$, the capital-weighted aggregate, because that is the object that nests to the macro β_t ; the population-median β is instead stable near one across waves (Figure 8). (b) Cross-sectional dispersion $P_{90}(\beta_i) - P_{10}(\beta_i)$ within each wave. (c) Wealth gradients: $\rho(\beta_i, \log a_i)$ (solid) and $\rho(z_i, \log a_i)$ (dashed) within each wave. (d) Capital-weighted $\hat{\beta}_{cap}$ by within-survey wealth quartile, SCF only. The low SCF point in 1995 in panels (a) and (d) is a list-sample composition outlier (a few list-sample households carry extreme capital weights), as in the note to Table 8; the median is unaffected. All panels share a common 1990–2020 horizon.

$\beta_{cap} = 0.92$; SCF Q4 has median $\beta = 0.92$ and $\beta_{cap} = 0.82$. Because the capital-weighted aggregate is dominated by Q4 (which holds $\sim 80\%$ of survey-aggregate wealth), the survey $\hat{\beta}_{cap}$ inherits Q4's value (~ 0.92 in PSID, ~ 0.82 in SCF); SCF's Q4 reaches deeper into the ultra-wealthy than PSID's, which is why the SCF aggregate sits lower. The bottom three quartiles look conventional: rapid accumulation at the bottom, β falling toward the macro level as wealth rises. The top quartile drives the macro–micro gap and the cross-survey difference in $\hat{\beta}_{cap}$.

Productivity-wealth sorting evolves differently across surveys. PSID's $\rho(z_i, \log a_i)$ rises from 0.31 in 2005 to 0.46 in 2019; SCF's declines from 0.30 in 1992 to 0.16 in 2019 (Figure I.15c, dashed lines). The divergence is informative. PSID conditions on working-age households with valid wages: among them, productivity sorting strengthens over time, confirming the macro paper's sorting accelerator at the household level. SCF includes the very

top of the wealth distribution that PSID misses; at that top, wealth is increasingly capital-gains and inheritance driven rather than wage driven, so the SCF z - a correlation weakens secularly. The MRCE's second sorting covariance $\text{cov}(e, a)$ at the macro level absorbs this top-decile decoupling.

Table I.7: Discount factor by wealth quartile (PSID and SCF, all waves pooled)

Quartile	median a	median Δa %/yr	median β	β_{cap}
<i>PSID 2005–2019 (biennial saving equation)</i>				
Q1 (poorest)	0.20	+41.7	1.32	1.59
Q2	1.25	+9.3	1.05	1.19
Q3	3.99	+2.4	0.98	1.04
Q4 (top)	15.30	−1.9	0.93	0.86
<i>SCF 1992–2019 (intratemporal FOC, cross-sectional)</i>				
Q1 (poorest)	0.54	—	1.14	1.10
Q2	5.17	—	1.01	1.01
Q3	23.93	—	0.98	0.97
Q4 (top)	433.87	—	0.93	0.82

Notes: Quartiles formed within each survey using sample wealth a_t . Median a and median Δa are unweighted within quartile (the SCF dealer-sample weights tiny at the top and the PSID weights roughly uniform; weighted medians within Q4 mask its dissaving). Median a in model units (1 unit $\approx$ \$38,545, 2021 dollars). Δa is the median annualized geometric biennial wealth growth rate $(a_{i,t+2}/a_{i,t})^{1/2} - 1$ (PSID only). $\beta_{\text{cap}} = E_w[\beta_i a_i R_i]/E_w[a_i R_i]$ within quartile, weighted.

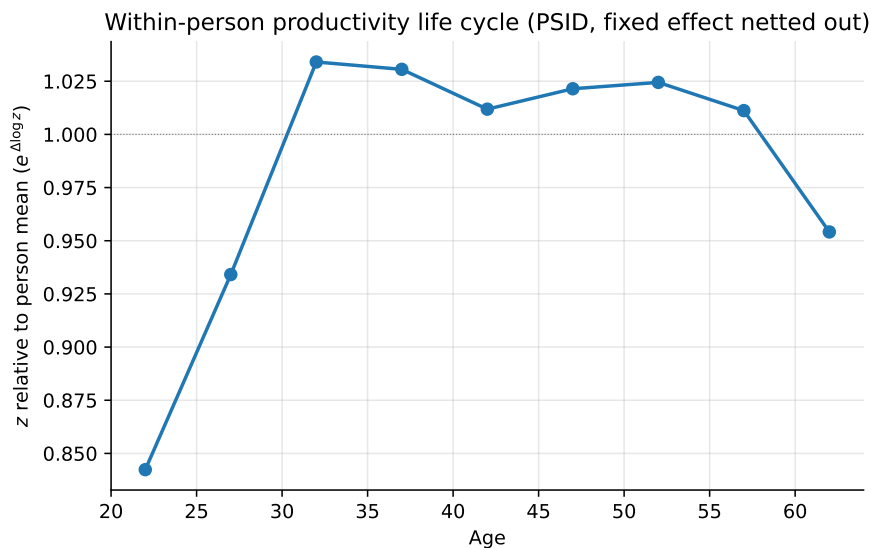
Table I.8: Age-conditional cross-parameter correlations (waves pooled within survey)

Age bin	n	$\rho(\beta, z)$	$\rho(\beta, \log a)$	$\rho(z, \log a)$
<i>PSID 2005–2019</i>				
[25, 29]	2,199	+0.08	+0.01	+0.44
[30, 39]	5,804	+0.04	−0.04	+0.49
[40, 49]	5,954	−0.01	−0.08	+0.44
[50, 59]	6,159	−0.06	−0.08	+0.39
[60, 64]	2,029	−0.01	−0.09	+0.37
<i>SCF 1992–2019</i>				
[25, 29]	10,008	−0.18	−0.41	+0.18
[30, 39]	30,285	−0.27	−0.52	+0.27
[40, 49]	39,851	−0.34	−0.54	+0.30
[50, 59]	38,192	−0.30	−0.48	+0.25
[60, 64]	12,524	−0.32	−0.41	+0.19
[65, 74]	10,779	−0.07	−0.10	+0.05

Notes: Within-bin Pearson correlations, weighted by survey weight, pooled across waves. Both PSID and SCF β clipped at $[0.5, 1.5]$.

Within-person productivity life cycle. The raw cross-sectional z profile mixes aging with composition: who is observed at work at each age changes over the life cycle, and at older ages the survey retains the high earners. Figure 1.16 removes this by netting out each person’s fixed effect in the PSID panel (persons observed in ≥ 2 waves; $\log z$ demeaned within person), leaving the pure aging profile. It rises from about 0.84 of the person mean at age 22 to a peak near 1.03 at age 32–37, holds near 1.02 through the early 50s, then declines to 0.95 by age 62. The shape matches the median cross-sectional profile in Figure 8, confirming that the hump is genuine aging and not a selection artifact.

Figure I.16: Within-person productivity life cycle (PSID, person fixed effect removed)



Notes: PSID persons observed in at least two waves. $\log z$ is demeaned within person, then averaged by 5-year age bin with survey weights; the plotted value is $e^{\Delta \log z}$, i.e. z relative to the person’s own lifetime mean. This isolates the within-person aging effect from the composition of who is observed at each age.

Robustness: what drives PSID’s dispersion and SCF’s clip. PSID’s wide β dispersion is a feature of the biennial saving identification, not of the underlying population. Re-identifying β on the *same* PSID households through the static labor FOC, the route SCF uses, collapses the weighted $P_{90} - P_{10}$ from 1.57 to 0.60, with the median unchanged at 1.01 and close to SCF’s own 0.36. The cross-survey difference in dispersion is therefore almost entirely the identification method, the noisy two-year wealth change versus the static margin, rather than a real difference in patience heterogeneity. The small residual gap is that PSID holds more wealth-poor and low-hours households than SCF, so a few more reach the clipped corners (14% versus 8% outside $[0.5, 1.5]$). The baseline therefore uses the static labor FOC

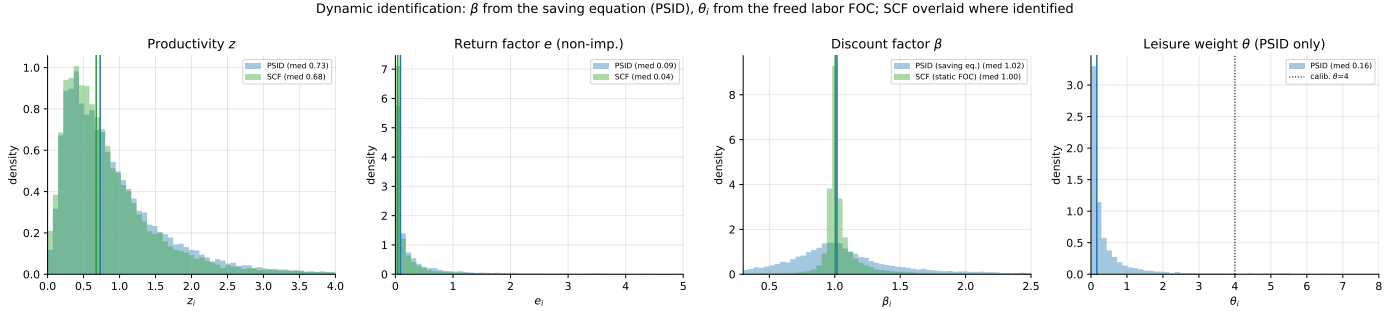
for β_i in both surveys, the route that makes them comparable. PSID’s panel additionally supports the dynamic saving-equation route, which we report here as a robustness check and use in the θ_i extension (Appendix G.8), where reading β_i off the wealth change frees the labor FOC to identify the leisure weight; the median and the capital-weighted aggregate, the objects we report, are insensitive to which route is used.

The SCF clip at $[0.5, 1.5]$ caps observations where the static FOC, which divides by wealth, returns uninformative extremes. The raw unclipped values run to absurd magnitudes ($P_{99} = 3.8$, with an upper tail past 4,000 and a negative lower tail), so the clip collapses nonsense onto the boundary rather than discarding interior values; 92% of the sample is interior. The two corners are economically distinct. The 4.7% clipped to 1.5 are full-time workers with near-zero net worth (median net worth $\sim \$5,000$, median hours 2,340, 97% in managerial or professional occupations): the small wealth denominator makes β explode upward. The 3.1% clipped to 0.5 are high earners, partly genuine (the SCF list sample reaches multi-million-dollar wages, which are not top-coded) and partly a low-hours artifact of the full-time-equivalent wage (about a third work fewer than 500 hours), both of which inflate z_i and push β down. SCF net worth is directly observed for every household and is not imputed, so the wealth-poor corner is real data; an hours floor that removes the low-hours artifact leaves the median profiles unchanged.

Parameter distributions under each identification scheme. Figures I.17 and I.18 show the parameter distributions under the two identification schemes, with PSID held to the same households across the two figures and SCF overlaid for comparison. Productivity z_i and the return factor e_i are recovered first, from the wage and capital income, so they are identical across the schemes and similar across the surveys (PSID and SCF medians 0.73 and 0.68 for z , 0.09 and 0.04 for e). The schemes differ only in the discount factor and in whether a leisure weight is recovered. Under the *dynamic* scheme (Figure I.17), the PSID β_i comes from the biennial saving equation and is wide ($P_{90} - P_{10} = 1.41$), while the SCF β_i , which can only be static, is a tight peak; the freed labor FOC additionally identifies a PSID leisure weight θ_i (Appendix G.8), strongly right-skewed and mostly well below the calibrated $\theta = 4$ (median 0.16; 55% of households fall outside the admissible range $q_i \in (0, 1)$ and are imputed to the calibrated value). SCF cannot identify θ_i , so that panel is PSID only. Under the *static* scheme (Figure I.18), both surveys identify β_i from the static labor FOC at fixed θ , and the two now coincide as tight peaks at one ($P_{90} - P_{10} = 0.49$ in PSID and 0.32 in SCF). Put together, the figures localize the cross-survey dispersion gap entirely in the identification method: the same static route makes the two surveys agree, while the biennial saving route is what spreads PSID out. All three parameters are also right-skewed, so the weighted mean

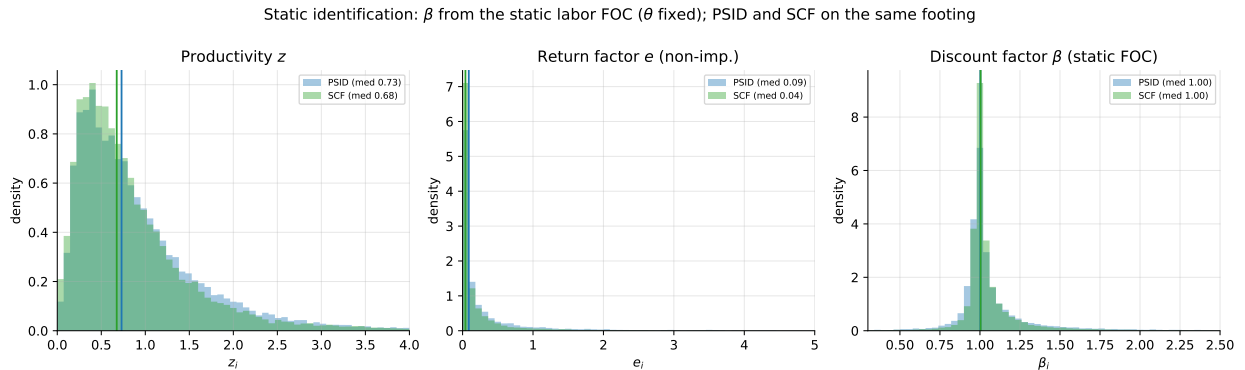
overstates the typical household (for the PSID saving-equation β , mean 1.30 against median 1.02); the lifecycle figures (Figure 8) therefore report survey-weighted medians.

Figure I.17: Parameter distributions, dynamic identification (β from the saving equation, θ_i from the freed labor FOC), SCF overlaid



Notes: Survey-weighted densities, waves pooled, PSID (blue) and SCF (green); vertical lines are weighted medians. z_i and e_i are identical to the static scheme (they are recovered before β). The β panel pairs the PSID saving-equation β with the SCF static-FOC β , which is the only route SCF admits. The θ_i panel is PSID only (SCF cannot identify it) and shows the 45% of households with an admissible $q_i \in (0, 1)$; the remaining 55% are imputed to the calibrated $\theta = 4$ (dotted line). Axes truncated for readability.

Figure I.18: Parameter distributions, static identification (β from the static labor FOC, θ fixed), PSID and SCF

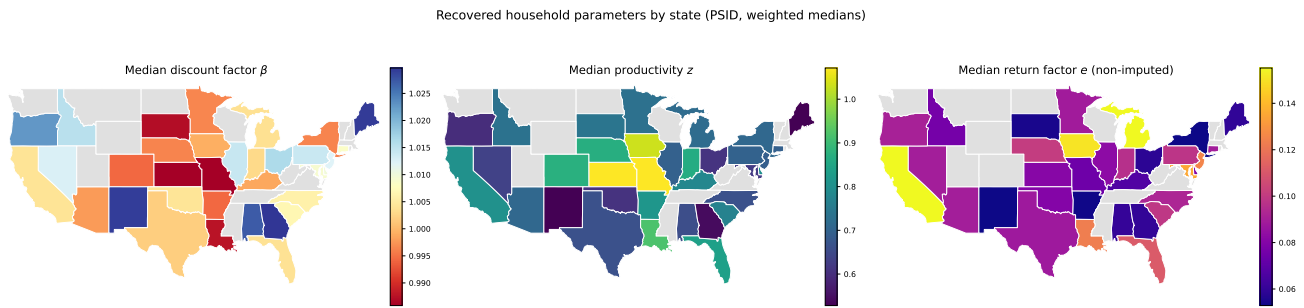


Notes: Survey-weighted densities, waves pooled, PSID (blue) and SCF (green); vertical lines are weighted medians. The PSID rows are the same households as Figure I.17. z_i and e_i are identical to the dynamic scheme. Under the common static identification the discount-factor panel is far tighter than its dynamic counterpart and the two surveys coincide, since the static FOC does not pass the two-year wealth-change noise into β . The PSID static β here is the unclipped reconstruction; in the baseline it is clipped to $[0.5, 1.5]$. Axes truncated for readability.

Geography of the recovered parameters. Because the inversion runs household by household, the recovered (z_i, e_i, β_i) can be mapped geographically. Figure I.19 plots their PSID weighted medians by state. Productivity z varies widely (state medians 0.41 to 1.24),

tracking regional labor markets, and the return factor e also varies, from 0.04 to 0.18 across states (among households with reported capital income), while the discount factor β is comparatively uniform (0.98 to 1.04): patience is far more geographically homogeneous than ability or returns. The z and β maps are negatively related across states, with high- z states tending to have lower median β , the geographic footprint of the within-state wealth gradient $\rho(\beta_i, \log a_i) < 0$ (Section 5.2.1): where productivity and wealth are higher, the wealthy dissave at lower β . These cross-state differences are descriptive, since they partly reflect wealth and labor composition rather than pure preference geography.

Figure I.19: Recovered household parameters by state (PSID, weighted medians)



Notes: PSID weighted-median discount factor β (left), productivity z (center), and non-imputed return factor e (right) by state, pooling all waves; **state** is FIPS-coded. Continental U.S. The β and z maps use states with ≥ 100 observations (36 states); the e map uses only households with reported capital income and states with ≥ 60 such observations (35 states), since the 59% of households with imputed $e = 1$ would otherwise dominate the median. States below the threshold are left blank. Color scales span the 5th–95th percentiles across states. Cross-state differences are descriptive and partly reflect wealth and labor composition.

References

- Bach, L., L. E. Calvet, and P. Sodini (2020). Rich pickings? Risk, return, and skill in household wealth. *American Economic Review* 110(9), 2703–2747.
- Bloom, N., M. Floetotto, N. Jaimovich, I. Saporta-Eksten, and S. J. Terry (2018). Really Uncertain Business Cycles. *Econometrica* 86(3), 1031–1065.
- Brock, W. A. and L. J. Mirman (1972). Optimal economic growth and uncertainty: The discounted case. *Journal of Economic Theory* 4(3), 479–513.
- Caballero, R. J. and M. L. Hammour (1994). The cleansing effect of recessions. *American Economic Review* 84(5), 1350–1368.
- Chari, V. V., P. J. Kehoe, and E. R. McGrattan (2007). Business cycle accounting. *Econometrica* 75(3), 781–836.
- Constantinides, G. M. and D. Duffie (1996). Asset pricing with heterogeneous consumers. *Journal of Political Economy* 104(2), 219–240.
- Fagereng, A., L. Guiso, D. Malacrino, and L. Pistaferri (2020). Heterogeneity and persistence in returns to wealth. *Econometrica* 88(1), 115–170.
- Kaplan, G., B. Moll, and G. L. Violante (2018). Monetary policy according to HANK. *American Economic Review* 108(3), 697–743.
- Kesten, H. (1973). Random difference equations and renewal theory for products of random matrices. *Acta Mathematica* 131, 207–248.
- Koop, G., M. H. Pesaran, and S. M. Potter (1996). Impulse Response Analysis in Nonlinear Multivariate Models. *Journal of Econometrics* 74(1), 119–147.
- Krusell, P. and A. Smith, Jr. (1998, October). Income and Wealth Heterogeneity in the Macroeconomy. *Journal of Political Economy* 106(5), 867–896.
- Piketty, T. (2014). *Capital in the Twenty-First Century*. Cambridge, MA: Harvard University Press.
- Saez, E. and G. Zucman (2016). Wealth inequality in the United States since 1913: Evidence from capitalized income tax data. *Quarterly Journal of Economics* 131(2), 519–578.
- Storesletten, K., C. I. Telmer, and A. Yaron (2004). Cyclical dynamics in idiosyncratic labor market risk. *Journal of Political Economy* 112(3), 695–717.